

Internet Appendix for “The Intramonth Momentum Cycle”

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This Internet Appendix contains supplementary material for “The Intramonth Momentum Cycle.” It includes: (IA.1) data construction and sample formation, (IA.2) holding period analysis, (IA.3) transaction cost decomposition, (IA.4) T+1 settlement falsification tests, (IA.5) sliding window analysis, (IA.6) within-month permutation test on the PreTOM window, (IA.7) tail trimming analysis, (IA.8) crash-day concentration methodology, (IA.9) international evidence methodology and full country-level results, (IA.10) subperiod PreTOM concentration analysis, (IA.11) window dressing falsification, (IA.12) dividend income flow pressure and payment-cycle detail, (IA.13) 52-week-high sort additional evidence, (IA.14) replication of Carhart (1997) extended through 2025 and the T+1 settlement difference-in-differences on UMD and loser funds, (IA.15) beta controls, (IA.16) the European T+3-to-T+2 settlement transition, (IA.17) the U.S. T+3-to-T+2 settlement transition and the aggregate fund-flow evidence around it, (IA.18) winner window effect, (IA.19) S&P 500 membership robustness, (IA.20) long-horizon reversion identifying the PreTOM-specific component, and (IA.21) additional figures.

IA.1 Data Construction and Sample Formation

The daily stock-level panel used in the main analysis is constructed as follows.

1. **CRSP input data.** We obtain the CRSP daily stock file from WRDS in Flat File

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Format 2.0 (CIZ). The file spans December 29, 1978 through December 31, 2025.¹ All WRDS inputs were downloaded in March 2026; WRDS data are revised periodically, so later downloads may yield slightly different raw data.

2. **CRSP sample restrictions.** We restrict the CRSP sample to U.S. common operating-company equities listed on the NYSE, AMEX, or Nasdaq, and retain only regular-way, active observations.²
3. **Trading-day panel construction.** Using the CRSP trading calendar, we expand each security’s panel so that all trading days between its first and last observed trading day are included. Trading days without an observed CRSP record remain in the panel as missing rows.
4. **Month-relative trading-day index.** Each trading day is indexed relative to the turn-of-the-month (TOM). Within each calendar month, the last trading day is labeled τ , the second-to-last $\tau-1$, and so on through $\tau-9$ for the tenth-to-last trading day. Trading days at the start of a calendar month are indexed forward from the preceding month’s TOM: $\tau+1$ is the first trading day of the calendar month, $\tau+2$ the second, and so on. The early-month controls $[\tau+5, \tau+8]$ used in the main paper are therefore the fifth through eighth trading days of the calendar month, counted from the preceding TOM.
5. **Momentum construction.** For each stock-month, we compute 12-2 momentum from monthly returns aggregated from daily CRSP observations:

$$M_{i,m} = \prod_{s=m-12}^{m-2} (1 + r_{i,s}) - 1.$$

The resulting stock-month signal is assigned to all daily observations in holding month m .

¹Observations from 1978–1979 are retained in order to construct momentum signals for the first valid holding months in 1980.

²In the CIZ format, these restrictions are implemented sequentially as follows: `ShareType='NS'`, `SecurityType='EQTY'`, `SecuritySubType='COM'`, `USIncFlg='Y'`, `IssuerType` in `{ACOR, CORP}`, `PrimaryExch` in `{N, A, Q}`, `ConditionalType='RW'`, and `TradingStatusFlg='A'`. Legacy CRSP share-code filters such as 10 and 11 are not available in the same way in CIZ, so we use the equivalent CIZ security-type and trading-status fields instead.

6. **Momentum validity.** A momentum window is treated as valid only if it contains no trading-calendar-added empty rows and no missing-return indications according to the CRSP variable `DlyRetMissFlg`.³ We then retain only observations with valid momentum and non-missing lagged market capitalization. The final analysis panel used in the paper contains 53,324,149 observations on 22,744 securities, spanning January 2, 1980 through December 31, 2025.
7. **Momentum deciles and portfolio weights.** We assign each stock-month to a momentum decile using Kenneth French’s NYSE 2–12 prior-return breakpoints, aligned to the month immediately preceding the holding month. All stock-months that survive the post-momentum filters are successfully matched to a breakpoint date and receive a non-missing decile assignment. For the portfolio analysis, weights are computed within each $(date, decile)$ cell using lagged market capitalization:

$$w_{i,t}^{(q)} = \frac{\text{MCAP}_{i,t-1}}{\sum_{j \in q(t)} \text{MCAP}_{j,t-1}},$$

where $q(t)$ denotes the set of securities in the relevant decile on date t , and $\text{MCAP}_{i,t-1}$ is security i ’s market capitalization on the previous trading day. These lagged market-cap weights are included in the final analysis panel.

8. **Additional daily characteristics.** We merge S&P 500 membership from the CRSP index constituents database. We also merge daily factors from Kenneth French’s data library (*Fama/French 3 Factors*), from which we use the daily risk-free rate (`RF`) and the daily market return (`Mkt`) in the analysis. In addition, we construct the bid-ask spread measure

$$\text{BAS}_{i,t} = \frac{\text{Ask}_{i,t} - \text{Bid}_{i,t}}{(\text{Ask}_{i,t} + \text{Bid}_{i,t})/2}.$$

The S&P 500 membership indicator is available for all observations; 5,460,281 daily rows correspond to index members and 47,863,868 correspond to non-members. The BAS variable is non-missing for 46,560,017 out of 53,324,149 observations, corresponding to 87.32% coverage.

³In the baseline specification, a day is admissible only if `DlyRetMissFlg` is either absent or equal to NA. Thus, windows are excluded if they contain any day with `DlyRetMissFlg` in $\{\text{MV}, \text{NS}, \text{NT}, \text{RA}, \text{GP}, \text{MP}, \text{DG}, \text{DM}, \text{DP}\}$.

9. **TAQ sell-pressure measures.** We merge daily sell-pressure measures derived from WRDS TAQ intraday indicator data into the CRSP panel using PERMNO-date. The TAQ data are available from September 10, 2003 through October 27, 2022, so coverage is nonzero only within this period. The merge is performed as a left merge, so the number of daily panel rows is unchanged. Within the full 1980–2025 panel, 17,861,277 observations have non-missing net selling pressure, corresponding to 33.50% of the sample. Restricting to the TAQ-available period, the panel contains 18,217,051 observations on 9,416 securities, of which 98.05% have non-missing sell-pressure measures.
10. **Final panel.** The final dataset is a daily PERMNO-date panel spanning January 2, 1980 through December 31, 2025. It contains CRSP returns and market-cap variables, momentum measures and momentum deciles, the month-relative trading-day index, lagged market-cap portfolio weights, S&P 500 membership, daily factor variables, bid-ask spreads, and TAQ-based sell-pressure measures.

Mutual-fund sample. For the monthly Carhart replication (Table 10, 1963–2025), we restrict to actively managed U.S. equity diversified funds via CRSP’s objective code (`crsp_obj_cd` LIKE 'ED%'), excluding ETF share classes (`et_flag` = 'F') and ETNs (`et_flag` = 'N'), as well as any fund flagged as an index fund under the CRSP categorical scheme (`index_fund_flag` ∈ {D, E, B, BMT}, covering pure index, enhanced index, index-based, and benchmark-matched products). Funds are sorted on January 1 of each year into deciles by the prior calendar year’s return and held for the next twelve months; decile portfolios are equal-weighted within each month, and holding-period monthly returns are winsorized at 1%/99% within each decile to limit the influence of extreme observations on the alpha estimates. No AUM threshold is applied. For the daily fund analysis (Sections 6.3 and 7), we apply the same universe filter and additionally require AUM of at least \$10 million because the daily-NAV sample is dominated by very small share classes that the AUM filter screens out without affecting the substantive conclusions.

IA.2 Holding Period Analysis

A related question is whether the PreTOM concentration depends on the choice of holding period. Our baseline uses a one-month holding period ($K = 1$), which is standard in the factor zoo literature but shorter than the 3–12 month holding periods emphasized in Jegadeesh and Titman (1993). Table IA.1 addresses this by varying K from 1 to 12 months while keeping the formation period fixed at 12 months with a one-month skip, using NYSE-breakpoint decile assignments throughout. For $K > 1$, we implement the overlapping-portfolio construction of Jegadeesh and Titman (1993): the strategy holds K cohorts simultaneously, each formed in a different past month and each held for K months with fixed composition, and takes the equal-weighted average of their returns on each trading day. This is the standard approach for extending holding periods in the momentum literature (Carhart, 1997; Jegadeesh and Titman, 2001). Standard errors use Newey-West with $K - 1$ lags to account for the autocorrelation induced by the overlapping cohorts.

The PreTOM component is statistically significant at the 1% level for every holding period examined, while the Rest component is never significant. By $K = 9$, cumulative Rest wealth falls below 1 so PreTOM wealth exceeds total strategy wealth. By $K = 12$, the Rest component is approximately zero in bps/month, and the PreTOM share reaches 139%: the six pre-month-end trading days generate more wealth than the full strategy, while the remaining trading days contribute essentially nothing. The PreTOM concentration is not a feature of short holding periods. If anything, it becomes more pronounced as K increases.

This monotonicity is consistent with the month-end cash-demand mechanism. The information-driven component of momentum (which operates uniformly across trading days) decays quickly as the formation signal becomes stale, while the flow-driven PreTOM component (which reflects a monthly institutional cycle) persists as long as funds continue to hold stale losers.

Table IA.1: PreTOM Concentration by Holding Period

	$K = 1$	$K = 3$	$K = 6$	$K = 9$	$K = 12$
<i>Panel A: Loser – Market (bps/month)</i>					
PreTOM	−56.8*** (−4.44)	−55.9*** (−4.76)	−51.8*** (−4.63)	−48.4*** (−4.34)	−44.4*** (−4.22)
Rest	−34.0 (−1.52)	−31.7 (−1.50)	−24.3 (−1.27)	−16.7 (−0.92)	−9.5 (−0.54)
<i>Panel B: Winner – Market (bps/month)</i>					
PreTOM	3.5 (0.45)	0.7 (0.10)	2.3 (0.32)	0.6 (0.09)	−1.1 (−0.15)
Rest	−0.7 (−0.06)	−6.5 (−0.53)	−8.3 (−0.70)	−9.5 (−0.81)	−12.9 (−1.16)
<i>Panel C: WML (bps/month)</i>					
PreTOM	61.4*** (3.59)	58.2*** (3.85)	55.6*** (3.86)	50.4*** (3.60)	44.4*** (3.38)
Rest	38.4 (1.37)	29.8 (1.12)	19.5 (0.78)	10.6 (0.44)	−0.4 (−0.02)
<i>Panel D: Wealth (\$1 invested, 1980–2025)</i>					
\$PreTOM	18.78	16.91	15.81	12.43	9.33
\$Rest	2.37	1.79	1.25	0.86	0.53
\$Total	44.46	30.28	19.76	10.71	4.93
PreTOM share	77%	83%	93%	105%	139%

Notes. Momentum portfolios use the French 12-month formation period with a one-month skip and NYSE breakpoints for decile assignment. The holding period K varies from 1 to 12 months. For $K > 1$, the strategy holds K overlapping portfolios simultaneously and takes the equal-weighted average of their returns. Panels A–C report mean monthly compounded returns in basis points; t -statistics are in parentheses and use Newey-West standard errors with $K - 1$ lags. Market-adjusted returns subtract the raw market return. Each panel compounds its own daily series independently to monthly returns. Because compounding is non-linear, Panel C (WML) does not equal Panel B (Winner – Market) minus Panel A (Loser – Market): the difference is a Jensen / convexity term that reflects realized daily-return volatility differentials across the legs. Panel D reports cumulative wealth from \$1 invested over 1980–2025. PreTOM share = $\log(\$PreTOM) / \log(\$Total) \times 100$. Sample: 1980–2025.

IA.3 Transaction Cost Decomposition

Novy-Marx and Velikov (2016) shows that momentum is among the most expensive anomalies to implement. We decompose the standard value-weighted momentum strategy into PreTOM and Rest components. Transaction costs are computed following Novy-Marx and Velikov (2016): each stock that changes momentum decile incurs a round-trip cost equal to one full quoted bid-ask spread (half-spread to liquidate the position plus half-spread to enter the replacement). Costs are value-weighted within each leg and summed over the long (winner) and short (loser) sides of the long-short.⁴

Table IA.2 reports the results. Over the full sample (1980–2025), gross WML averages +91.3 bps per month, of which +59.7 bps (65%) accrues during PreTOM. Monthly-rebalanced costs of 79.9 bps per month leave a net return of +11.5 bps. A PreTOM-only strategy that enters and exits each month faces a full round-trip cost of 317.1 bps, making it unviable in the full sample due to pre-decimalization spreads. Post-decimalization (2001–2025), quoted spreads shrink dramatically (losers 24.4 bps, winners 14.8 bps), reducing the full round-trip cost to 39.2 bps. The PreTOM-only strategy earns +54.1 bps gross and +14.9 bps net of costs. The Rest strategy is negative both gross (−26.9 bps) and net (−66.1 bps). In modern markets, the entire net WML return originates in PreTOM, with the rest-of-month component net negative. The cost assigned to Rest-only is conservative. Rest-only remains invested across the calendar-month boundary and therefore incurs an additional partial rebalance each month, since decile assignments change between months. Incorporating this additional turnover would further reduce Rest-only’s net return relative to PreTOM-only; the asymmetry we document is therefore understated by the symmetric round-trip treatment.

⁴Our cost measure is conservative. Novy-Marx and Velikov (2016) uses the Hasbrouck (2009) effective spread, which is typically smaller than the quoted spread. Monthly turnover averages 21.0% for losers and 26.3% for winners.

Table IA.2: Transaction Cost Decomposition of Monthly Momentum Returns

	(1) Gross (bps/mo)	(2) TC (bps/mo)	(3) Net (bps/mo)
<i>Panel A: Full sample (1980–2025)</i>			
Buy-and-hold WML	+91.3	79.9	+11.5 (0.35)
PreTOM component	+59.7*** (3.60)		
Rest component	+32.4 (1.12)		
PreTOM-only strategy	+59.7	317.1	−257.5 (−11.34)
Rest-only strategy	+32.4	317.1	−284.8 (−8.99)
<i>Panel B: Post-decimalization (2001–2025)</i>			
Buy-and-hold WML	+25.1	10.1	+15.0 (0.29)
PreTOM component	+54.1** (2.07)		
Rest component	−26.9 (−0.59)		
PreTOM-only strategy	+54.1	39.2	+14.9 (0.56)
Rest-only strategy	−26.9	39.2	−66.1 (−1.43)

Notes. The table decomposes monthly returns of the standard value-weighted WML strategy (long decile 10, short decile 1, rebalanced monthly using fixed momentum decile breakpoints from CRSP) into PreTOM ($[\tau-9, \tau-4]$, where τ is the last trading day of the month) and rest-of-month components. Daily value-weighted returns within each window are compounded to monthly frequency. Transaction costs are computed following Novy-Marx and Velikov (2016) using quoted bid-ask spreads. “Buy-and-hold” charges one full quoted spread for each stock that changes momentum decile (monthly turnover: losers 21.0%, winners 26.3%). “PreTOM-only” and “Rest-only” strategies additionally incur a full round-trip spread on all portfolio stocks each month, since the investor enters and exits the entire long-short position within the month; Rest-only further bears a partial decile-rebalancing cost at each month boundary, which is not separately accounted for in the table. Post-decimalization, mean quoted spreads are 24.4 bps (losers) and 14.8 bps (winners), making the PreTOM-only strategy viable net of costs. t -statistics are in parentheses, placed beneath the number being tested. ** $p < 0.05$, *** $p < 0.01$.

IA.4 T+1 Settlement Falsification Tests

Three checks confirm that the T+1 settlement result is specific to the boundary days the mechanism predicts and is not the artifact of a slow pre-existing drift. First, *placebo days*: running the identical $\tau-4$ vs. $\tau-3$ DiD but shifted to $\tau-6$ vs. $\tau-7$, two days well inside both the T+2 and T+1 selling windows where the mechanism predicts no relocation of selling pressure, produces a DiD of -2.3 ($t = -0.08$). Second, *placebo dates*: the same $\tau-4$ vs. $\tau-3$ comparison using fake event dates spread across 1999–2015 (May 28 of 1999, 2005, 2011, 2013, and 2015), each well removed from major market disruptions, returns DiDs ranging from -13.4 to $+3.5$ with $|t| \leq 1.19$. All falsifications are indistinguishable from zero, in contrast to the $+84.7$ ($t = 2.80$) for the actual May 2024 reform estimated under this same no-control falsification specification (the controls-augmented difference-in-differences reported in the main text is $+85.9$, $t = 2.68$). Table IA.3 reports the full results. Third, the *event study* below shows that the within-month $\tau-4$ minus $\tau-3$ differential has no pre-trend in the years before the reform and lifts only after May 2024, supporting the parallel-trends assumption underlying the DiD.

Table IA.3: T+1 Settlement: Falsification Tests

Test	DiD	t -stat	p -value	Prediction
Real: Losers, $T-4$ vs. $T-3$, May 2024	84.72	2.80	0.005	Nonzero
<i>Placebo days (losers, May 2024):</i>				
$T-6$ vs. $T-7$	-2.33	-0.08	0.936	Zero
<i>Placebo dates (losers, $T-4$ vs. $T-3$):</i>				
May 28, 2015	3.54	0.19	0.848	Zero
May 28, 2013	3.08	0.19	0.849	Zero
May 28, 2011	3.01	0.20	0.838	Zero
May 28, 2005	-10.05	-0.79	0.432	Zero
May 28, 1999	-13.41	-1.19	0.233	Zero

Notes: Falsification tests for the T+1 settlement DiD reported in the main paper. All tests use the same specification: market-adjusted VW loser returns regressed on a day indicator, a post-event indicator, and their interaction, with robust standard errors. Placebo days shift the comparison to $T-6$ vs. $T-7$, two days interior to both the T+2 and T+1 selling windows where no shift is predicted. Placebo dates use fake event dates when no regulatory change occurred. All falsification interactions are small and statistically insignificant. Sample: 1980–2025.

Event-study pre-trend test. The pooled DiD restricts the data to a single pre/post split at May 28, 2024. A pre-trend in the within-month $\tau-4$ minus $\tau-3$ differential before that date would invalidate the DiD even if the placebo tests above were clean. We test this directly by replacing the T1Reform_t dummy variable with a vector of 12-month event-year bins anchored at May. Bin 0 covers May 2024–April 2025, bin -1 (the omitted reference) covers May 2023–April 2024, bin -2 covers May 2022–April 2023, and so on through bin -8 . All months from 1980 through April 2016 are pooled into a single “deep pre” bin. May 2024 is excluded throughout, matching the main-text spec. On the subsample $[\tau-4, \tau-3]$, we estimate

$$r_t = \alpha + \sum_k \alpha_k \mathbf{1}[\text{bin}_t = k] + \delta \mathbf{1}_{[\tau-4]} + \sum_{k \neq -1} \beta_k \mathbf{1}[\text{bin}_t = k] \times \mathbf{1}_{[\tau-4]} + \varepsilon_t,$$

where r_t is the daily market-adjusted value-weighted loser return in basis points. Each β_k is the within-month $\tau-4$ minus $\tau-3$ differential in event-year bin k , relative to the year just before the reform.

Figure IA.1 plots the eleven coefficients with 95% confidence intervals. The eight pre-event coefficients are scattered around zero, none significant at the 5% level and with no monotone drift toward May 2024; the joint test that all eight equal zero cannot be rejected ($F(8, 1080) = 0.59$, $p = 0.78$). Both post-event bins lift to similar positive magnitudes: +121.3 basis points for May 2024–April 2025 ($t = 2.00$, $p = 0.05$) and +111.1 for the partial bin May 2025–December 2025 ($t = 1.81$, $p = 0.07$), each on the order of the pooled DiD reported in the main paper. The flat pre-period, the discontinuous post-period jump, and the persistence of the effect into the second post-event bin together support the parallel-trends assumption.

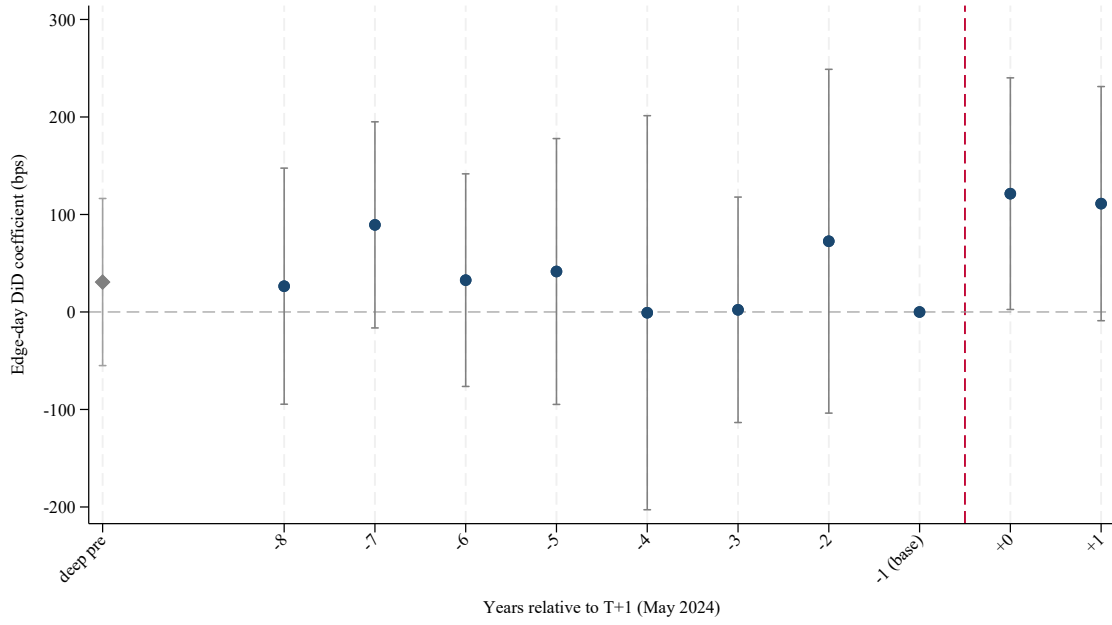


Figure IA.1: Event study of the $\tau-4$ vs. $\tau-3$ DiD. Each point is the within-month $\tau-4$ minus $\tau-3$ differential in market-adjusted value-weighted loser returns (basis points), estimated for the indicated 12-month bin relative to the omitted bin -1 (May 2023–April 2024). Bins are anchored at May, so bin 0 is May 2024–April 2025 and bin -2 is May 2022–April 2023. The deep-pre bin pools 1980 through April 2016. Whiskers are 95% confidence intervals from heteroskedasticity-robust standard errors. May 2024 is excluded. Sample: 1980–2025.

IA.5 Sliding Window Analysis

A natural concern is that the PreTOM window $[\tau-9, \tau-4]$ was selected ex post to maximize the effect. We address this by sliding a six-day window across the second half of the trading month and computing the raw mean VW loser market-adjusted return (loser – market) within each window. We restrict to windows starting on or after trading day 7 (approximately mid-month) to ensure that all days within each window belong to the same monthly portfolio assignment.

The mean loser market-adjusted return is near zero for mid-month windows, declines monotonically as the window shifts toward month-end, and troughs at PreTOM $([\tau-9, \tau-4])$: -11.32 basis points per day ($t = -3.53$). No other window starting position produces a comparably negative estimate. The decline into PreTOM and partial recovery at month-end are exactly what the month-end cash-demand mechanism predicts.

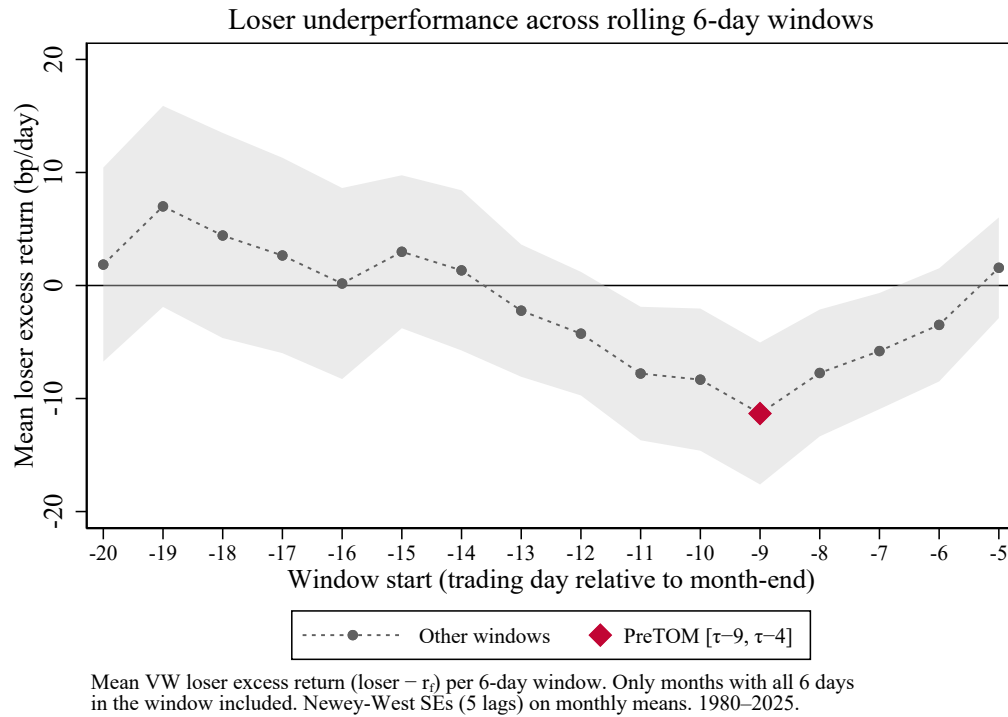


Figure IA.2: Sliding six-day window analysis. Each point plots the mean daily VW loser market-adjusted return (loser - market) for a six-day window starting at the indicated trading day. The trough occurs at PreTOM ($[\tau-9, \tau-4]$). Sample: 1980–2025.

IA.6 Within-Month Permutation Test on the PreTOM Window

To address the concern that the $[\tau-9, \tau-4]$ window is a post-hoc selection from the many possible six-day windows in a trading month, we perform a within-month permutation test on the daily series of value-weighted D1 loser market-adjusted returns (1980–2025).

For each calendar month independently, we assign a placebo τ by sampling one trading day uniformly at random from the set of trading days that allow $[\tau-9, \tau-4]$ to fall entirely within the month. Trading-day indices are recomputed relative to the placebo anchor, and the days at placebo-relative position $[\tau-9, \tau-4]$ form placebo-PreTOM. We compute the placebo statistic

$$\hat{\theta}^{\text{plac}} = \bar{r}_{t \in \text{placebo-PreTOM}}^{\text{D1, mkt-adj}} - \bar{r}_{t \notin \text{placebo-PreTOM}}^{\text{D1, mkt-adj}}, \quad (\text{IA.1})$$

pooled across all months. The procedure is repeated 10,000 times. The one-sided lower-tail p -value is the fraction of placebo draws with $\hat{\theta}^{\text{plac}} \leq \hat{\theta}$, where $\hat{\theta} = -7.12$ bps/day is the observed PreTOM-minus-Rest mean difference.

Table IA.4 reports the results. The observed estimate sits in the 1.2nd percentile of the null distribution ($p = 0.012$). A placebo test on D5 (middle-decile) market-adjusted returns produces $\hat{\theta} = -0.37$ bps/day at the 24.9th percentile ($p = 0.25$), confirming that the test discriminates between losers and non-losers rather than picking up generic calendar structure.

The D1 null distribution is centered slightly below zero (mean = -1.36 bps). This offset is mechanical: the within-month constraint forces month-end trading days ($\tau-3$ to $\tau-0$) to remain in placebo Rest in every draw. These days carry the post-PreTOM reversal documented in Section 4.3 ($\beta_3 = +3.36$ bps/day on the loser leg) and therefore bias placebo Rest’s mean upward. The test is conservative: the observed estimate sits in the 1.2nd percentile despite this downward shift in the null. The D5 placebo, which lacks comparable month-end loser dynamics, has a null mean essentially at zero ($+0.22$ bps), confirming the offset is loser-specific and not an artifact of the randomization scheme.

Figure IA.3 plots the D1 null distribution against the observed estimate.

Table IA.4: Within-Month Permutation Test for the PreTOM Window

	D1 (losers)	D5 (placebo)
<i>Panel A: Observed estimate and inference</i>		
$\hat{\theta}$ (bps/day)	−7.12	−0.37
Percentile in null	1.2nd	24.9th
One-sided p -value	0.012	0.249
<i>Panel B: Null distribution statistics</i>		
Null mean	−1.36	+0.22
Null sd	2.58	0.87
1st percentile	−7.24	−1.79
5th percentile	−5.62	−1.19
50th percentile	−1.39	+0.22
95th percentile	+2.88	+1.66
99th percentile	+4.63	+2.25
Skewness	+0.02	+0.00
Excess kurtosis	−0.04	+0.04
Number of draws	10,000	10,000

Notes. Within-month permutation test described in Internet Appendix Section IA.6. The observed statistic $\hat{\theta}$ is the mean D1 (or D5) value-weighted market-adjusted return on $[\tau-9, \tau-4]$ days minus the mean on all other days, pooled across 1980–2025. For each draw, the placebo τ is sampled within each month from trading days that allow $[\tau-9, \tau-4]$ to fall within the month; the placebo statistic is computed with these placebo labels. The D1 null mean is slightly negative because the within-month constraint excludes month-end days from placebo PreTOM in every draw; those days carry the post-PreTOM reversal of Section 4.3. The D5 placebo confirms the offset is loser-specific.

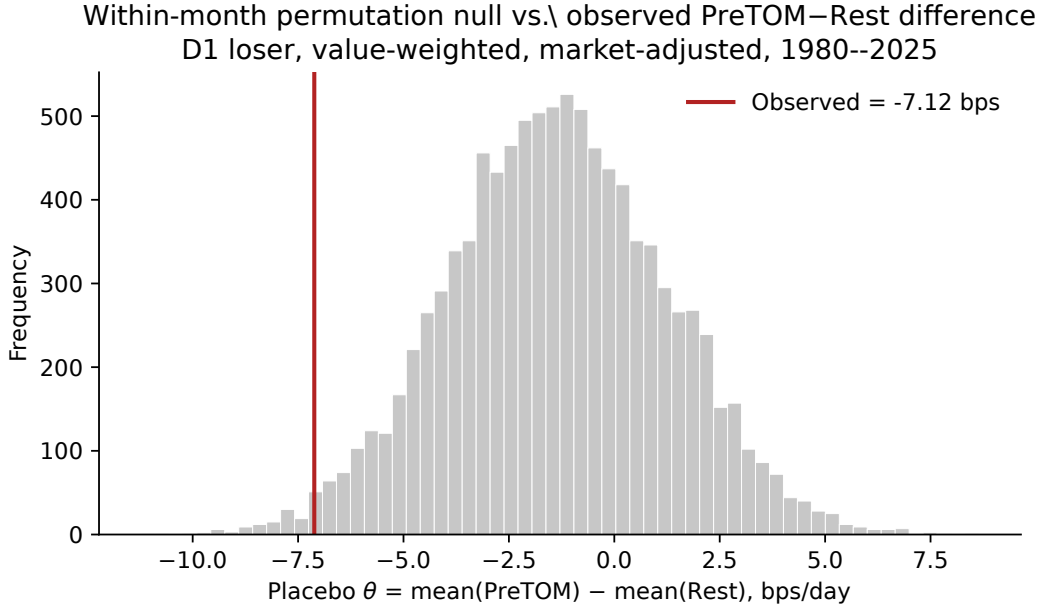


Figure IA.3: Empirical null distribution of the D1 loser PreTOM-minus-Rest mean difference under a within-month permutation test (10,000 draws). For each month and each draw, a placebo τ is sampled uniformly at random from trading days that allow $[\tau-9, \tau-4]$ to fall within the month; placebo PreTOM is the set of days at placebo-relative position $[\tau-9, \tau-4]$, and the placebo statistic is the mean over placebo PreTOM minus the mean over the remaining days. The vertical line marks the observed estimate of -7.12 bps/day, which sits in the 1.2nd percentile of the null distribution (one-sided $p = 0.012$). The null distribution is centered slightly below zero because the within-month constraint excludes month-end days ($\tau-3$ to $\tau-0$) from placebo PreTOM in every draw; those days carry the post-PreTOM reversal documented in Section 4.3, raising placebo Rest’s mean. A parallel test on D5 (middle decile) yields a null mean of $+0.22$ bps and $p = 0.25$ (Table IA.4). Value-weighted, market-adjusted, 1980–2025.

IA.7 Tail Trimming Analysis

A key concern is whether PreTOM profitability is structural or driven by a few extreme days. We address this by progressively trimming the worst days from each window's return distribution and recomputing the mean. Figure IA.4 shows the result. PreTOM mean WML is +10.2 bps untrimmed and rises to +17.2 bps after removing the worst 1% of days, confirming that the effect is structural and not driven by outliers. In contrast, month-start mean WML flips from -4.4 bps to +6.2 bps after the same 1% trim: approximately 15 extreme days drive the entire negative month-start average. This asymmetry confirms that PreTOM profits and month-start crashes have distinct statistical signatures: PreTOM is a persistent daily effect, while month-start negativity is concentrated in rare tail events.

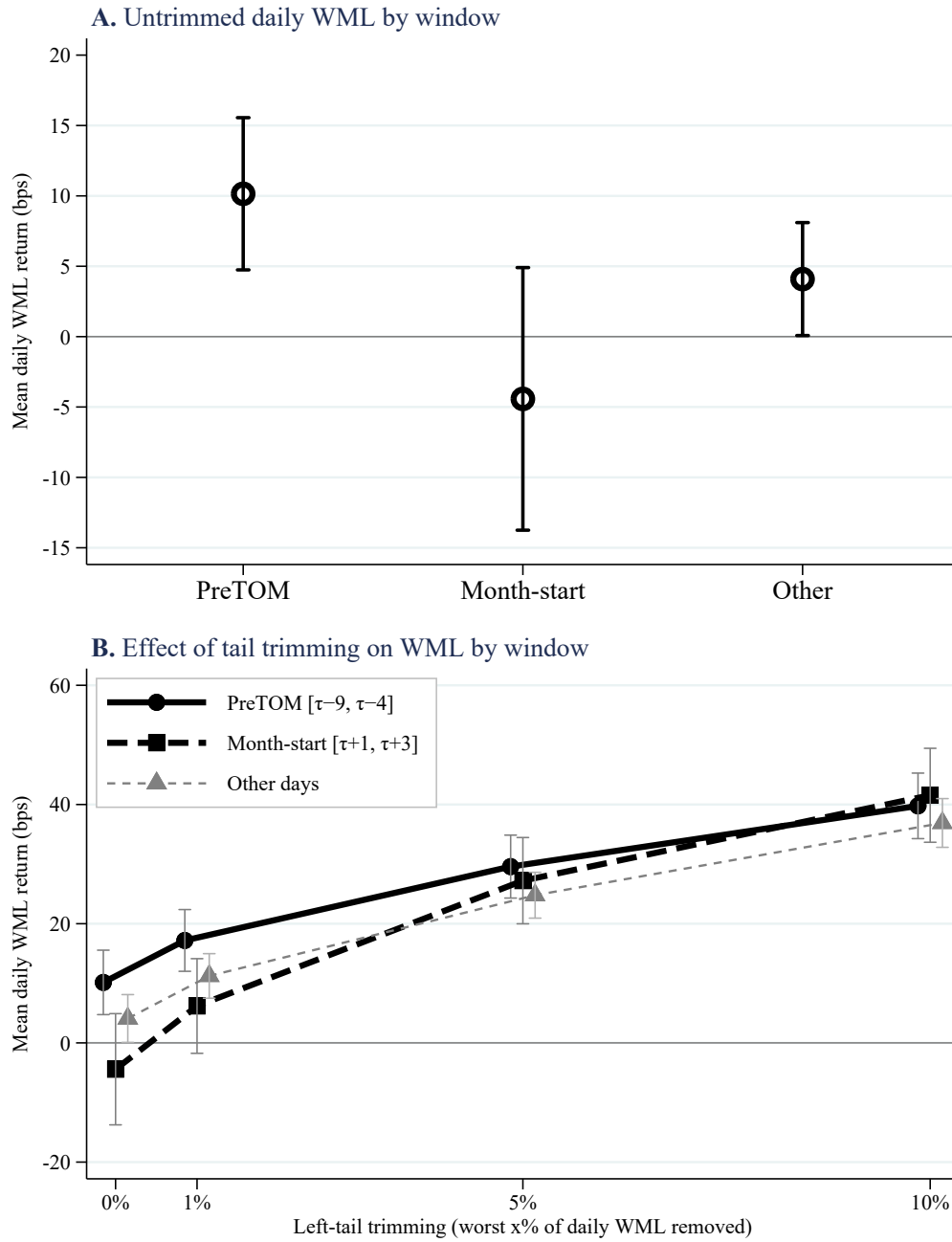


Figure IA.4: Mean daily WML return by calendar window after progressively trimming the worst days from each window's distribution. Panel A: pre-month-end ($[\tau-9, \tau-4]$). Panel B: month-start ($[\tau+1, \tau+3]$). PreTOM mean WML rises from +10.2 to +17.2 bps after 1% trimming (structural), while month-start flips from -4.4 to +6.2 bps (driven entirely by ~ 15 tail events). Sample: 1980–2025, value-weighted.

IA.8 Crash-Day Concentration Methodology

This appendix details the calendar-share construction and proportions z -test summarized in Section 8 of the main paper.

Crash-day classification. We classify a trading day as a crash day if the value-weighted WML return falls below -200 basis points. Robustness at -100 and -300 basis-point thresholds is reported in the figures.

Comparison universes. To prevent the PreTOM and month-start windows from contaminating each other, we compare each window’s crash share against its calendar share within a comparison universe that excludes the other window. For PreTOM, the universe is non-month-start trading days (PreTOM days plus other days, excluding $[\tau+1, \tau+3]$). For month-start, the universe is non-PreTOM trading days (month-start days plus other days, excluding $[\tau-9, \tau-4]$). The calendar share is the fraction of trading days in the universe that fall in the window of interest. Under the null that crashes are spread proportionally across the calendar, the observed share equals the calendar share.

Test statistic. For each window w and threshold, let $p_{\text{obs}} = n_{\text{crash},w}/n_{\text{crash}}$ be the observed share of crash days in window w and p_{exp} be the calendar share. The standard one-sample proportions z -statistic is

$$z = \frac{p_{\text{obs}} - p_{\text{exp}}}{\sqrt{p_{\text{exp}}(1 - p_{\text{exp}})/n_{\text{crash}}}},$$

two-sided p -values from the standard normal.

Excluded episodes. The robustness reported in the body excludes 11 months that contain famous momentum-crash episodes: January 2001, October–November 2001, and November 2002 (the Daniel–Moskowitz 2016 cluster); March–May 2009; and February–May 2020 (COVID-19 dislocation).

IA.9 International Evidence: Methodology

This appendix details the construction of the international momentum decile portfolios used in the international evidence presented in the main paper.

Sample. Daily stock returns are from Compustat Global (`comp.g_secd`) for 19 developed markets: Australia, Austria, Belgium, Denmark, Finland, France, Germany, Hong Kong, Ireland, Italy, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, and the United Kingdom. Country is identified by the issuer-of-record (`loc`) field rather than the listing exchange. The sample begins in 1990 for most countries and in 1993 for Hong Kong and Switzerland, where Compustat coverage is thin in the early 1990s. We exclude Japan, where momentum returns are approximately zero over our sample period (Fama and French, 2012; Asness et al., 2013), and Canada, for which Compustat Global coverage is limited (Canadian common stocks are concentrated in the CRSP North-America universe).

Stock universe and cleaning. We retain common stocks only (`tpci='0'`). The total daily return is constructed from price (`prccd`), the cumulative split-adjustment factor (`ajexdi`), and the cumulative dividend reinvestment factor (`trfd`):

$$r_{i,t} = \frac{p_{i,t}/a_{i,t} \cdot \tau_{i,t}}{p_{i,t-1}/a_{i,t-1} \cdot \tau_{i,t-1}} - 1.$$

We drop observations with missing return or market capitalization, non-positive market capitalization, and returns outside $[-0.99, 10]$ to remove obvious data errors. Market capitalization is `prccd` $\times$ `cshoc`.

Monthly aggregation. Daily returns are compounded to monthly returns within each stock-month. A stock-month is included only if it contains at least 15 daily observations; this rules out partially-traded months that would otherwise distort the momentum signal.

Momentum signal. For each formation month m , the signal is the cumulative log return over the eleven months $m-12$ through $m-2$, requiring all eleven monthly returns to be observed. Deciles are formed at the end of m and held in calendar month $m+1$. This follows

the standard 12-2 convention used in Asness et al. (2013) and the broader international momentum literature.

Decile assignments. Within each country-month, we rank stocks into momentum deciles using *size-conditional* breakpoints: decile cutoffs are computed from stocks above the within-country median market capitalization at the prior month-end, then all stocks (including small-cap) are assigned to deciles based on those cutoffs. This avoids decile assignments being driven by micro-cap noise. For seven countries with thin large-cap coverage we pool breakpoints within a broader region: small-Europe (Austria, Belgium, Finland, Ireland, Portugal) and small-Asia-Pacific (New Zealand, Singapore). All other countries (Australia, Denmark, France, Germany, Hong Kong, Italy, the Netherlands, Norway, Spain, Sweden, Switzerland, the United Kingdom) compute breakpoints from their own large-cap stocks each month. We additionally require at least 20 above-median stocks in the relevant breakpoint group (country or pooled region) in a given month to assign deciles; country-months that fall below this threshold are dropped from the panel. Decile assignments are fixed within each calendar month.

Returns and market adjustment. Value-weighted portfolio returns use prior-month-end market capitalization as weights. The regional market excess return is taken from Kenneth French’s regional factor files: *Europe 3 Factors Daily* for the European countries, *Asia Pacific ex Japan 3 Factors Daily* for Australia, Hong Kong, New Zealand, and Singapore. We subtract the relevant regional market excess return to obtain market-adjusted returns. The dependent variable in equation (10) is the day- t value-weighted loser-decile return in country c in excess of its regional market return, scaled to basis points.

Code reference. The full pipeline is implemented in `_intl_build_v3.py` in the replication archive. The output of this pipeline is the country-day panel `compustat_intl_pooled_v3.csv` on which Table 9 regressions are estimated.

Full country tables. Tables IA.5 and IA.6 report PreTOM, Rest, and Difference estimates for all 19 countries individually, for both legs (Losers – Market and Winners – Market). Countries are sorted by the loser-leg Difference column.

Table IA.5: International Evidence: All Countries — Losers — Market (bps/day)

Country	PreTOM		Rest of Month		Difference	
	Mean	<i>t</i>	Mean	<i>t</i>	Mean	<i>t</i>
Ireland	−7.36	(−1.34)	4.95	(1.17)	−12.32*	(−1.80)
Norway	−12.10***	(−3.26)	−0.73	(−0.32)	−11.37***	(−2.63)
Netherlands	−7.93*	(−1.96)	2.65	(1.14)	−10.57**	(−2.31)
Hong Kong	−9.15***	(−2.73)	−0.21	(−0.09)	−8.94**	(−2.21)
Sweden	−4.62	(−1.19)	3.68*	(1.71)	−8.30*	(−1.90)
Finland	−5.98	(−1.57)	1.92	(0.65)	−7.90	(−1.61)
Australia	−8.23***	(−3.28)	−1.56	(−0.84)	−6.67**	(−2.24)
Switzerland	−4.59	(−0.98)	1.86	(0.78)	−6.45	(−1.19)
Belgium	−6.55**	(−2.08)	−1.78	(−0.61)	−4.77	(−1.11)
France	−4.85*	(−1.77)	−0.18	(−0.08)	−4.67	(−1.35)
Spain	−3.80	(−1.19)	0.35	(0.21)	−4.15	(−1.14)
New Zealand	−2.91	(−0.67)	0.87	(0.30)	−3.78	(−0.66)
Denmark	−3.79	(−1.18)	−1.09	(−0.51)	−2.70	(−0.68)
Germany	−5.05*	(−1.81)	−2.75	(−1.26)	−2.30	(−0.63)
UK	−3.09	(−1.06)	−1.35	(−0.65)	−1.74	(−0.45)
Portugal	−3.22	(−0.84)	−3.41	(−1.29)	0.19	(0.04)
Italy	−2.78	(−0.91)	−5.65*	(−1.93)	2.87	(0.69)
Singapore	6.16**	(2.09)	2.69	(1.33)	3.47	(1.01)
Austria	−0.57	(−0.17)	−5.63**	(−2.14)	5.06	(1.21)
<i>Pooled</i>	−4.48***	(−5.71)	−0.26	(−0.47)	−4.21***	(−4.38)

Notes: Daily VW portfolio returns from Compustat Global, 1990–2025 (HKG/CHE from 1993). Momentum signal is the cumulative log return over months −12 through −2, requiring all eleven months observed (Asness et al., 2013 convention). NYSE-style breakpoints (above-median mcap). Countries with thin large-cap coverage use regional breakpoints (small-Europe: AUT, BEL, FIN, IRL, PRT; small-Asia-Pacific: NZL, SGP). Market-adjusted using Kenneth French regional factors. Sorted by Difference. Individual *t*-statistics clustered by month; pooled includes country FE in the Difference column and clusters by country-month.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.6: International Evidence: All Countries — Winners — Market (bps/day)

Country	PreTOM		Rest of Month		Difference	
	Mean	<i>t</i>	Mean	<i>t</i>	Mean	<i>t</i>
Ireland	1.26	(0.28)	1.31	(0.44)	−0.05	(−0.01)
Norway	2.37	(0.63)	5.58***	(2.65)	−3.21	(−0.74)
Netherlands	5.89	(1.46)	1.43	(0.70)	4.46	(0.98)
Hong Kong	−3.21	(−0.86)	1.26	(0.59)	−4.47	(−1.01)
Sweden	5.06*	(1.73)	4.51***	(2.70)	0.55	(0.15)
Finland	3.19	(0.95)	2.76	(1.18)	0.43	(0.10)
Australia	5.99**	(2.27)	4.83***	(2.82)	1.16	(0.35)
Switzerland	5.95**	(2.17)	1.58	(0.85)	4.37	(1.27)
Belgium	−5.79*	(−1.84)	3.71	(1.47)	−9.50**	(−2.38)
France	1.92	(0.81)	2.15	(1.18)	−0.23	(−0.07)
Spain	5.51*	(1.94)	3.65**	(2.05)	1.87	(0.55)
New Zealand	4.21	(0.93)	1.81	(0.73)	2.39	(0.46)
Denmark	5.15	(1.54)	4.47**	(2.22)	0.68	(0.16)
Germany	6.34*	(1.65)	2.53	(1.29)	3.81	(1.02)
UK	0.94	(0.31)	2.73	(1.63)	−1.79	(−0.50)
Portugal	6.68	(1.38)	4.75*	(1.86)	1.93	(0.35)
Italy	4.55*	(1.86)	3.00*	(1.75)	1.55	(0.52)
Singapore	4.08	(1.16)	1.98	(0.97)	2.10	(0.49)
Austria	5.37	(1.63)	1.42	(0.52)	3.94	(0.93)
<i>Pooled</i>	3.22***	(4.25)	2.87***	(6.16)	0.34	(0.38)

Notes: Same specification as Table IA.5. Countries sorted in the same order as the losers table for comparability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.10 Subperiod PreTOM Concentration: VW versus EW

Table IA.7 reports daily PreTOM and Rest mean returns for D1 (loser–Mkt) and WML by historical subperiod under both value-weighting (Panel A) and equal-weighting (Panel B). PreTOM days are 29% of the trading month. If daily returns were generated uniformly across the month, PreTOM and Rest mean returns would coincide up to sampling error; any systematic gap between them reflects a calendar-time concentration in the underlying return process.

PreTOM losses dominate Rest-of-month returns in every subperiod from 1927 onward, but the gap widens substantially after 1962, coinciding with the scaling of institutional ownership in U.S. equities.

Table IA.7: Subperiod Analysis: PreTOM versus Rest, D1 (Loser – Mkt) and WML

<i>Panel A: D1 (Loser – Mkt) returns (bps/day)</i>					
Period	Value-weighted		Equal-weighted		Inst. Own.
	PreTOM	Rest	PreTOM	Rest	
1927–1940	–5.06 (–1.00)	–4.50 (–1.61)	19.69 (3.57)	23.50 (7.46)	< 10%
1940–1962	–2.35 (–1.77)	–2.09 (–2.37)	2.22 (1.45)	4.33 (3.95)	< 10%
1962–1980	–8.01*** (–5.47)	–2.56 (–2.62)	–1.66*** (–1.15)	4.60 (4.33)	15–30%
1980–2002	–9.65* (–4.72)	–4.14 (–3.16)	8.46 (4.07)	11.38 (8.50)	30–55%
2002–2025	–6.46** (–1.85)	2.15 (0.94)	3.08 (1.21)	3.66 (2.09)	55–72%
<i>Panel B: WML (Winner – Loser) returns (bps/day)</i>					
Period	Value-weighted		Equal-weighted		Inst. Own.
	PreTOM	Rest	PreTOM	Rest	
1927–1940	2.25 (0.34)	9.10 (2.32)	–13.59 (–2.38)	–12.06 (–3.63)	< 10%
1940–1962	5.75 (2.92)	4.85 (3.75)	3.58 (1.88)	1.10 (0.86)	< 10%
1962–1980	10.74 (4.91)	7.92 (5.56)	6.80* (4.29)	2.40 (2.10)	15–30%
1980–2002	10.24 (3.40)	7.06 (3.77)	–6.01 (–3.28)	–4.66 (–3.81)	30–55%
2002–2025	10.06** (2.21)	–1.91 (–0.64)	1.08 (0.42)	–3.12 (–1.73)	55–72%

Notes. Daily mean returns in basis points per day, by historical subperiod and weighting scheme. Panel A: D1 (loser – Mkt) portfolio return. Panel B: WML (winner – loser) excess return. PreTOM is the trading-day window $[T-9, T-4]$ before month-end; Rest is all other trading days. PreTOM and Rest level cells report mean and robust t -statistic; the diff t -statistic that drives the stars uses Newey-West (21 lags) on the joint regression of the daily series on PreTOM. Asterisks on PreTOM cells denote significance of the PreTOM versus Rest difference, not significance of the level. Aggregate institutional ownership figures are approximate ranges from Blume and Keim (2012) and the NYSE Fact Book. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.11 Window Dressing Falsification

Brown (2017) documents that momentum returns concentrate in the third month of calendar quarters, attributing the pattern to tax-loss selling and window dressing by delegated managers.

If window dressing is the primary driver of the pre-month-end loser effect, the pattern should amplify in December and at quarter-ends. Our evidence points in the opposite direction. The PreTOM effect is present every month, with no quarter-end amplification, and is not absorbed by December tax-driven selling.

Three tests address the window dressing and tax-loss alternatives directly, all reported in Table IA.8. First, restricting the sample to non-quarter-end months, where window dressing is least relevant (column 1), the Loser $\times$ PreTOM coefficient is -7.5 bps ($t = -2.66$), virtually identical to the full-sample estimate. Second, restricting the sample to non-December months, where tax-loss harvesting is least relevant (column 2), the Loser $\times$ PreTOM coefficient is -7.03 bps ($t = -2.86$), again virtually identical to the full-sample estimate of -7.15 bps. Third, interacting Loser $\times$ PreTOM with a quarter-end indicator (column 3) produces a small, insignificant triple interaction: there is no systematic quarter-end amplification, consistent with a monthly liquidity cycle rather than quarterly reporting incentives.

Table IA.8: Window Dressing and Tax-Loss Falsification

	(1) Non-QE months	(2) Non-December months	(3) Full sample, QE interaction	(4) Full sample baseline
Loser	4.132*** (2.68)	2.756** (2.05)	4.039*** (2.62)	2.609** (2.04)
Loser $\times$ PreTOM	-7.523*** (-2.66)	-7.027*** (-2.86)	-7.496*** (-2.65)	-7.151*** (-3.08)
Loser $\times$ QtrEnd			-4.249 (-1.48)	
Loser $\times$ PreTOM $\times$ QtrEnd			0.990 (0.18)	
Constant	-0.017 (-0.12)	-0.078 (-0.63)	-0.162 (-1.41)	-0.162 (-1.39)
Fixed effects	Stock, Date	Stock, Date	Stock, Date	Stock, Date
Observations	35,417,562	48,853,202	53,324,145	53,324,145
Within R^2	0.0000	0.0000	0.0000	0.0000

Notes. Dependent variable is the daily market-adjusted return ($r_{i,t} - r_t^m$, in excess of the value-weighted CRSP market return), in basis points. Returns are value-weighted using lagged market-capitalization weights. Column (1) restricts the sample to non-quarter-end months (Jan, Feb, Apr, May, Jul, Aug, Oct, Nov); window dressing would be least relevant in these months, yet the Loser $\times$ PreTOM coefficient is essentially identical to the full-sample estimate. Column (2) restricts the sample to non-December months; tax-loss harvesting is most concentrated in December, and excluding it leaves the Loser $\times$ PreTOM coefficient essentially unchanged. Column (3) estimates the full sample with quarter-end interactions; QtrEnd equals one in March, June, September, December. The triple interaction Loser $\times$ PreTOM $\times$ QtrEnd is small and statistically indistinguishable from zero, indicating no quarter-end amplification. Column (4) reproduces the full-sample baseline from Table 2 of the main paper for reference. All specifications include stock and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: 1980–2025.

IA.12 Dividend Income: Flow Pressure and the Payment Cycle

This section reports the within-quarter payment-cycle detail and the flow-pressure evidence behind the dividend-income channel summarized in Section 4.2.2 of the main paper. A dividend payment delivers a cash buffer that drains over the months until the next payment. We examine whether PreTOM selling pressure on a stock varies with its position in the payment cycle, and whether any such pattern is concentrated in losers rather than winners. We do not derive the within-quarter pattern as a strict implication of the dispensability mechanism, since stock-specific dividend timing need not translate into stock-specific selling pressure when institutions manage cash at the portfolio level. We present the within-cycle estimates as descriptive evidence consistent with a within-cycle modulation of cash needs at the stock level.

We test this within D1 and D10 quarterly payers. With $\text{Month}_{i,t} \in \{1, 2, 3\}$ indexing the position of day t within stock i 's dividend quarter ($m = 1$: dividend month; $m = 2$: one month after; $m = 3$: two months after), we estimate

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha \text{Loser}_{i,t} + \sum_{m=1}^3 \beta_m^L \text{Loser}_{i,t} \times \text{PreTOM}_t \times \mathbf{1}[\text{Month}_{i,t} = m] \\ & + \sum_{m=1}^2 \beta_m^W \text{PreTOM}_t \times \mathbf{1}[\text{Month}_{i,t} = m] + \sum_{m=1}^2 \lambda_m \text{Loser}_{i,t} \times \mathbf{1}[\text{Month}_{i,t} = m] \\ & + \sum_{m=1}^2 \theta_m \mathbf{1}[\text{Month}_{i,t} = m] + \mu_i + \delta_t + \varepsilon_{i,t}. \end{aligned}$$

Month 3 is the omitted reference for the lower-order Month and Loser $\times$ Month interactions. Standard errors are two-way clustered by stock and date. The standalone $\text{Loser}_{i,t}$ captures the within-stock time-varying component of decile membership; stock fixed effects μ_i absorb only its stock-level mean. Date fixed effects δ_t absorb the standalone PreTOM_t indicator. Table IA.9 reports the estimates in Panel B (returns) and Panel C (TAQ).⁵

⁵Coefficients β_m^L report the loser-specific Loser $\times$ PreTOM $\times$ Month effect in each dividend cycle month rather than as differentials from Month 3; the saturated triple-interaction structure identifies all three m jointly with the $\alpha \text{Loser}_{i,t}$ standalone. Winner coefficients β_m^W for $m \in \{1, 2\}$ are PreTOM $\times$ Month effects for non-losers relative to the Month 3 reference. The TAQ analog replaces $r_{i,t} - r_t^m$ with $\text{NSP}_{i,t}$ and relabels $\beta \rightarrow \gamma$; the right-hand side is identical.

Both empirical patterns appear in the TAQ data. Panel C of Table IA.9 reports the loser coefficients rising monotonically across the cycle: $\gamma_1^L = +0.0018$ ($t = 0.70$), $\gamma_2^L = +0.0051$ ($t = 1.84$), $\gamma_3^L = +0.0057$ ($t = 2.12$). The winner coefficients are statistically flat: $\gamma_1^W = +0.0001$ ($t = 0.06$), $\gamma_2^W = +0.0016$ ($t = 0.77$); the joint test $\gamma_1^W = \gamma_2^W = 0$ does not reject ($F = 0.38$, $p = 0.68$). The dividend-cycle pattern is loser-specific in the data.

Panel B (returns) is directionally consistent but underpowered to detect cross-month differences. Loser coefficients are $\beta_1^L = -6.3$ bps/day ($t = -1.32$), $\beta_2^L = -10.6$ ($t = -1.98$), $\beta_3^L = -7.7$ ($t = -1.57$); the joint Wald does not reject equality across months ($F = 0.31$, $p = 0.73$). Winner coefficients are flat ($\beta_1^W = -4.4$, $\beta_2^W = -0.9$). Trade direction is more informative than returns at this sample size, consistent with the broader pattern that institutional selling shows more directly in TAQ flows than in daily returns.

Figure IA.5 provides direct flow-pressure evidence consistent with the dividend-income channel. During PreTOM, flow pressure is more negative for non-dividend payers than for dividend payers, and the gap is widest in the loser deciles: non-dividend payers in the bottom decile face flow pressure of -6.5 , compared with -1.4 (fund momentum, Panel A) and -0.1 (52-week-high distance, Panel B) for dividend payers in the same decile. The pattern appears under both groupings, consistent with investors selling securities with lower foregone dividend income when raising cash. We present this as cross-sectional evidence on which securities are selected for sale during the PreTOM cash-raising window, not as an independent timing test.

Table IA.9: Dividend Income and PreTOM Selling Pressure

<i>Panel A: Dividend Payer Composition</i>			
	D1 Losers	D10 Winners	
Non-payer	84.1%	67.5%	
Quarterly payer	8.7%	23.2%	
Other (semi-annual, annual, monthly)	7.2%	9.3%	
<i>Panel B: Within-Quarter Returns (Quarterly Payers, D1 ∪ D10)</i>			
	Month 1 (just paid)	Month 2 (1 mo after)	Month 3 (about to pay)
Winners: PreTOM × Month _m	−4.35 (−1.48)	−0.87 (−0.28)	0 (ref.)
Losers: Loser × PreTOM × Month _m	−6.28 (−1.32)	−10.56** (−1.98)	−7.74 (−1.57)
Observations	2,449,358		
<i>Panel C: Within-Quarter TAQ Net Selling Pressure (Quarterly Payers, D1 ∪ D10)</i>			
	Month 1 (just paid)	Month 2 (1 mo after)	Month 3 (about to pay)
Winners: PreTOM × Month _m	+0.0001 (0.06)	+0.0016 (0.77)	0 (ref.)
Losers: Loser × PreTOM × Month _m	+0.0018 (0.70)	+0.0051* (1.84)	+0.0057** (2.12)
Observations	768,018		

Notes. Panel A reports the share of stock-month observations by dividend frequency. Panels B and C restrict to quarterly dividend payers within D1 (loser) and D10 (winner) momentum portfolios. Month 1 is the dividend-payment month; Month 2 is one month after payment; Month 3 is the month before the next payment. Both panels report the triple-interaction specification described in Internet Appendix Section IA.12; Panel C uses $\text{NSP}_{i,t}$ as the LHS in place of $\text{ExRet}_{i,t}$. The “Winners” row is β_m^W , the $\text{PreTOM} \times \text{Month}_m$ coefficient (D10 baseline) for $m \in \{1, 2\}$ relative to Month 3. The “Losers” row is β_m^L , the $\text{Loser} \times \text{PreTOM} \times \text{Month}_m$ triple interaction, saturated across $m \in \{1, 2, 3\}$. All specifications include stock and date fixed effects; standard errors are two-way clustered by stock and date. Panel B is value-weighted by lagged market capitalization. Momentum deciles use NYSE breakpoints; assignments held constant within each calendar month. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

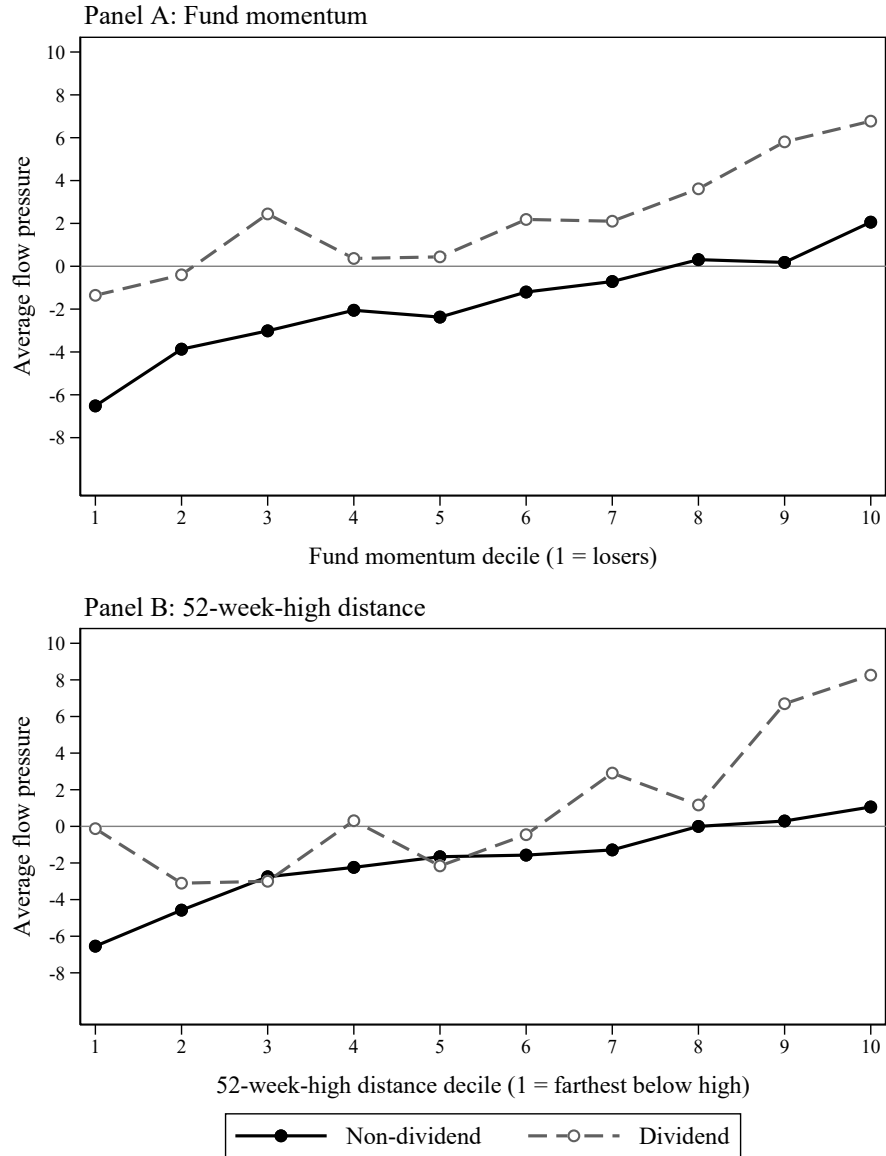


Figure IA.5: Flow pressure by dividend status during PreTOM. Each line plots average mutual-fund-flow pressure during the PreTOM window $[\tau-9, \tau-4]$ by decile, separately for non-dividend payers (solid line, filled markers) and dividend payers (dashed line, hollow markers). Panel A groups by fund momentum (decile 1 = losers); Panel B groups by distance from the 52-week high (decile 1 = farthest below the high). Lower values indicate stronger selling pressure. Flow pressure is constructed from mutual fund flows mapped to portfolio holdings.

IA.13 52-Week-High Sort: Additional Evidence

This section reports supporting figures for the 52-week-high evidence discussed in Section 4.2.4 of the main paper. Figure IA.6 compares the share of non-paying firms across deciles under the two sorts. Figure IA.7 reports cumulative \$1 growth on long-short portfolios sorted by the two characteristics over 1980–2025. Table IA.10 and Figure IA.8 report the PreTOM fund-flow double sort by fund momentum and distance from the 52-week high.

Non-payer share by decile: 12-2 momentum vs 52-week high (1980-2025)

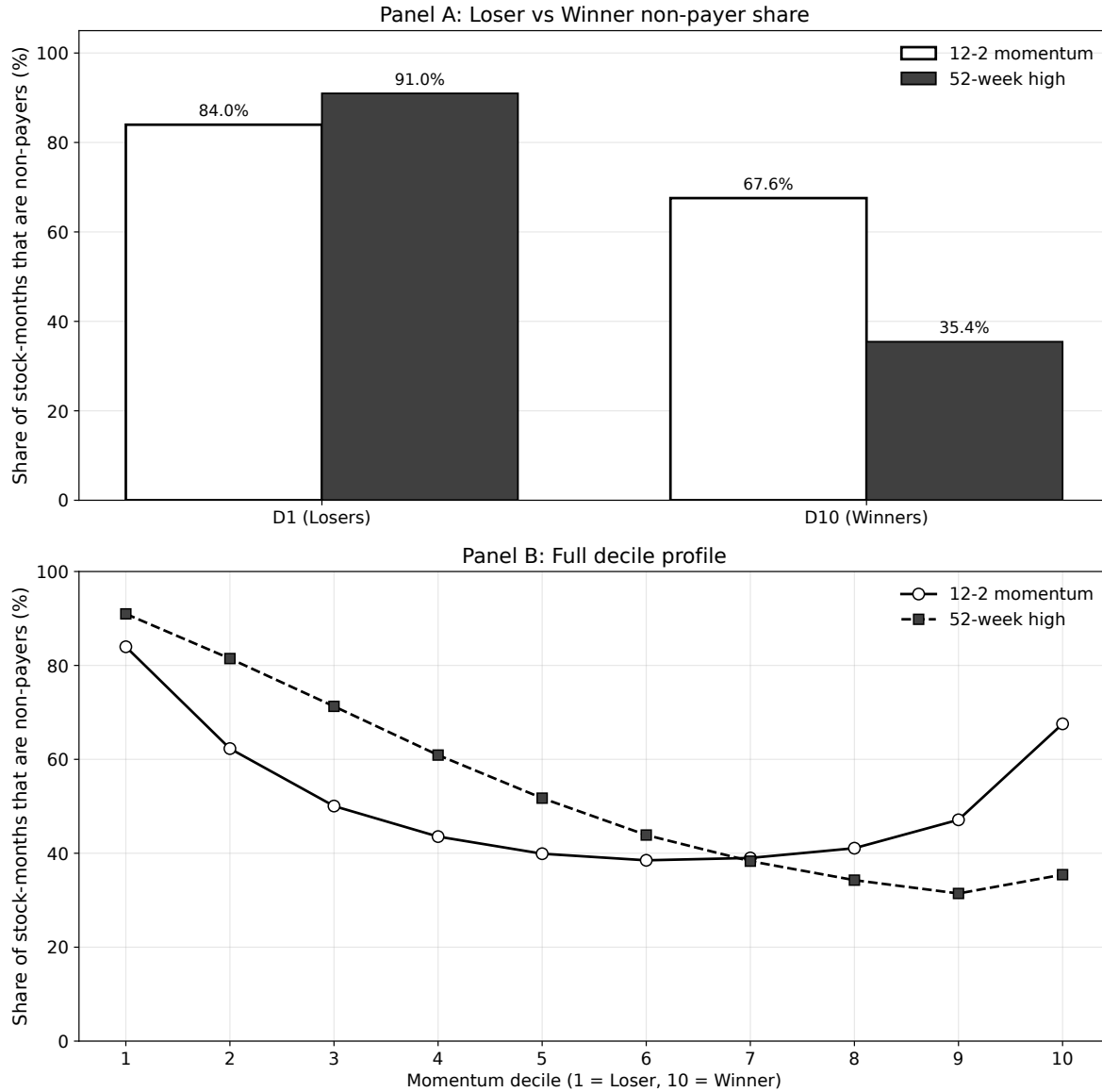


Figure IA.6: Share of non-paying stock-months by momentum decile under two sorts. Panel A: D1 (Losers) and D10 (Winners) shown as grouped bars, with 12-2 momentum (white bars with black border) and 52-week-high distance (dark gray bars). Panel B: full decile profile, 12-2 momentum (solid line with hollow markers) and 52-week-high distance (dashed line with filled markers). Both sorts are formed at the end of each month from all NYSE, AMEX, and NASDAQ common stocks in CRSP; assignments are held constant within the calendar month. Dividend-payer status is taken from CRSP distributions. Sample: 1980–2025.

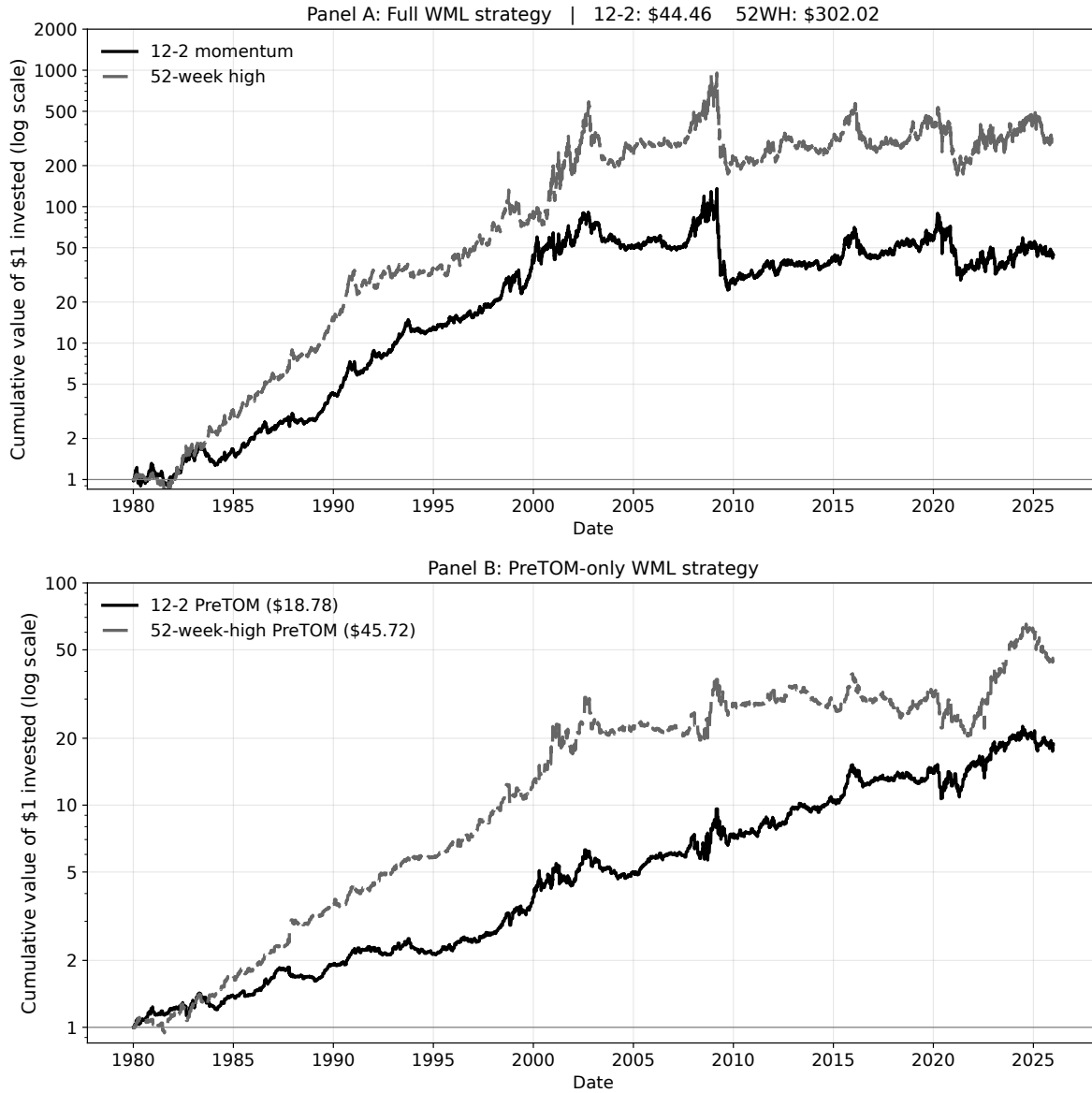


Figure IA.7: Cumulative \$1 invested in a long-short (winner – loser) portfolio over 1980–2025 under two sorts: standard 12-2 momentum (solid lines) and 52-week-high distance (dashed lines). Panel A: Full WML strategy, compounding daily WML returns every trading day. Panel B: window-conditional strategies, with PreTOM-only (in black) compounding only on days $[\tau-9, \tau-4]$ and Rest-of-month (in gray) compounding only on days outside that window; the strategy holds cash on the complementary days. Both panels are log-scaled on the y -axis with a common range. Value-weighted, fixed monthly decile assignments. End-of-sample wealth labels are shown in each legend entry.

Table IA.10 reports average mutual-fund-flow pressure during PreTOM in a double sort of funds by momentum decile and by distance-to-52-week-high decile. Outflow pressure is strongest where the two sorts agree. Loser-momentum funds farthest from their 52-week high face flow pressure of -9.7 , the most negative cell in the table, while loser-momentum funds near their 52-week high face only -2.1 . Figure IA.8 plots these corner profiles. The momentum ranking alone does not determine which funds face the outflows; distance from the 52-week high separates them within momentum ranks.

Table IA.10: PreTOM Flow Pressure: Fund Momentum $\times$ 52-Week-High Distance Double Sort

Momentum decile	Distance-to-52-week-high decile										Avg.
	1 (far)	2	3	4	5	6	7	8	9	10 (at high)	
1 (losers)	-9.72	-5.82	-5.74	-5.27	-3.98	-3.01	-3.32	-2.59	-1.02	-2.15	-6.33
2	-7.35	-5.74	-4.43	-2.84	-3.93	-0.45	-1.87	-1.61	-0.20	+0.63	-3.67
3	-5.60	-5.17	-2.39	-1.59	-1.40	-5.65	-2.10	-1.92	+3.70	+1.20	-2.46
4	-4.44	-5.01	-5.28	-3.39	-1.54	-1.17	-1.68	-1.27	+3.94	+0.24	-2.12
5	-2.83	-5.55	-1.81	-2.89	-1.41	-1.78	+1.33	-0.67	-4.29	-3.64	-2.08
6	-3.76	-3.85	-1.05	-1.10	-0.08	-0.37	-1.50	-0.08	+0.24	+0.89	-0.84
7	-1.62	-2.00	-4.60	+1.18	-2.36	-2.41	-0.98	+1.29	+2.02	+1.83	-0.47
8	-0.08	-2.23	+2.88	-1.35	-1.85	+0.98	-0.51	-1.39	+1.38	+5.90	+0.58
9	-1.62	-1.11	-2.03	-1.48	-1.09	-2.75	+2.64	+3.18	+2.58	+1.59	+0.51
10 (winners)	-1.44	-2.42	+1.84	+0.52	-1.92	+2.00	+2.11	+3.16	+4.08	+5.74	+2.16
Avg.	-5.77	-4.42	-2.68	-1.91	-1.82	-1.42	-0.38	+0.19	+1.52	+2.25	

Notes. Each cell reports average mutual-fund-flow pressure during the PreTOM window $[\tau-9, \tau-4]$ (τ is the last trading day of the month) for funds in the given fund-momentum and distance-to-52-week-high decile. Flow pressure is the sum of fund flows divided by the sum of lagged assets under management within each cell. Decile 1 of the momentum sort contains loser funds; decile 1 of the 52-week-high sort contains funds farthest below their 52-week high. Row and column averages are observation-weighted. Cell observation counts range from 4,337 to 45,149 fund-month observations. Lower values indicate stronger outflow pressure.

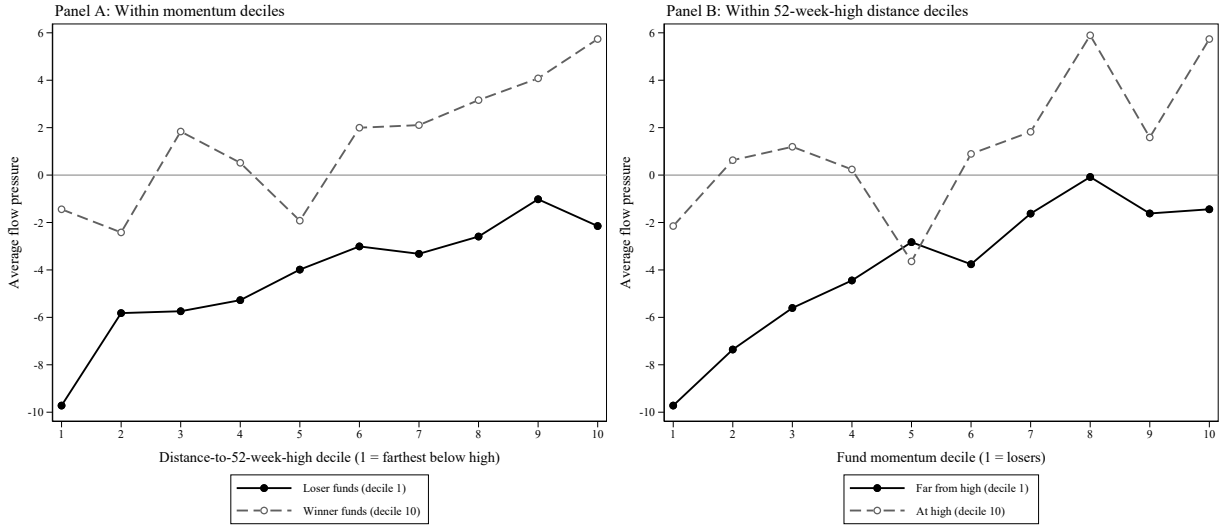


Figure IA.8: PreTOM flow pressure in the fund momentum $\times$ 52-week-high distance double sort. Panel A plots average flow pressure during PreTOM across distance-to-52-week-high deciles for loser-momentum funds (solid line, filled markers) and winner-momentum funds (dashed line, hollow markers). Panel B plots the same across fund momentum deciles for funds farthest from (solid) and at (dashed) their 52-week high. Lower values indicate stronger outflow pressure. Values correspond to Table IA.10.

IA.14 Replicating Carhart at Monthly Frequency

This section reports the monthly replication of Carhart (1997) extended through 2025, referenced in Section 7 of the main paper. We sort actively managed equity funds into deciles each year on prior 12-month return and regress each decile’s equal-weighted monthly excess return on the four Fama–French–Carhart factors:

$$R_m^{Dk} - r_m^f = \alpha^{Dk} + \beta^{Dk} \text{MKT-}rf_m + s^{Dk} \text{SMB}_m + h^{Dk} \text{HML}_m + u^{Dk} \text{UMD}_m + \varepsilon_m, \quad (\text{IA.2})$$

where R_m^{Dk} is the equal-weighted month- m return on decile Dk . Table IA.11 reports both the four-factor specification and the three-factor version (omitting UMD). Loser funds (D1) load on UMD with β between -0.17 and -0.26 in every subperiod from 1963 to 2025, and underperform the three-factor benchmark by -0.35% per month ($t = -4.07$).⁶ Winner funds (D10) outperform by $+0.37\%$ per month in 1963–1993 ($t = 3.84$) but the effect fades: $+0.06\%$ per month in 1994–2025 ($t = 0.59$) and mildly negative -0.14% per month in 2011–2025 ($t = -1.11$).

The daily-frequency settlement evidence referenced in Section 7.4 of the main paper appears in Table IA.12. Panel A reports the T+1 settlement difference-in-differences on the Fama–French daily UMD factor; Panel B reports the same specification on the equal-weighted D1 loser-fund portfolio.

⁶Choi and Zhao (2021) documents that full-month fund persistence disappears post-1994, attributing the disappearance to lower returns to favorable styles, less favorable style tilts among winners, and increased style-adjusted underperformance by winning funds. Our table is consistent with this characterization: the $D10 - D1$ FF3 spread attenuates to -0.03 ($t = -0.12$) in 2011–2025, down from $+0.74$ ($t = 4.54$) in 1963–1993. The attenuation reflects movement on both legs: D1 underperformance moderates from -0.36 to -0.11 across the endpoints, while D10’s FF3 alpha shifts from $+0.37$ in 1963–1993, to $+0.06$ in 1994–2025, and to -0.14 in 2011–2025 (consistent with the increased style-adjusted underperformance Choi and Zhao (2021) documents for winners).

Table IA.11: Carhart (1997) Replication: Monthly Fund Decile Alphas Across Subperiods
% per month, equal-weighted

	(1) 1963–1993 (Carhart)	(2) 1994–2025 (Post)	(3) 2011–2025 (Modern)	(4) Full (1963–2025)
<i>3-Factor alpha (FF3):</i>				
D1 (past losers)	−0.36*** (−3.02)	−0.32** (−2.46)	−0.11 (−0.86)	−0.35*** (−4.07)
D10 (past winners)	0.37*** (3.84)	0.06 (0.59)	−0.14 (−1.11)	0.23*** (3.23)
D10–D1	0.74*** (4.54)	0.37* (1.90)	−0.03 (−0.12)	0.58*** (4.58)
<i>4-Factor alpha (FF3 + UMD):</i>				
D1 alpha	−0.20* (−1.67)	−0.11 (−1.07)	0.01 (0.09)	−0.16** (−2.07)
D1 UMD β	−0.17*** (−4.40)	−0.26*** (−7.20)	−0.23*** (−3.99)	−0.22*** (−7.60)
D10–D1 alpha	0.32** (1.98)	−0.02 (−0.13)	−0.27 (−1.32)	0.19 (1.61)

Notes. Four-factor (Fama–French–Carhart) and three-factor (FF3) regression estimates of equation (IA.2) for CRSP equity-diversified mutual funds (ED, excluding index funds and ETFs), sorted on January 1 of each year into deciles by the prior calendar year’s return and held for the next twelve months; equal-weighted; monthly returns winsorized at 1%/99% within each decile. The D10–D1 row reports the alpha from a separately estimated regression of the monthly D10–D1 spread portfolio return on the factors, which restricts to months in which both deciles are non-empty; it therefore need not equal the algebraic difference of the D10 and D1 alphas, which use all months where the respective decile is non-empty. *t*-statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.12: Settlement Change Difference-in-Differences: UMD and Loser Funds
Panel A: T+1 settlement DiD on the Fama–French daily UMD factor (1980–2025)

	Coefficient
Intercept (early-month $[\tau+5, \tau+8]$ base)	2.39 (1.18)
$\mathbf{1}_{[\tau-4]}$	2.87 (0.75)
$\mathbf{1}_{[\tau-3]}$	0.42 (0.10)
T1Reform _t	−1.62 (−0.13)
$\theta_{\tau-4}$ (T1Reform _t $\times$ $\mathbf{1}_{[\tau-4]}$)	−42.63 (−1.61)
$\theta_{\tau-3}$ (T1Reform _t $\times$ $\mathbf{1}_{[\tau-3]}$)	35.45 (1.46)
DiD ($\theta_{\tau-3} - \theta_{\tau-4}$)	78.08** (2.48)
Observations	3,303

Notes. Boundary-day DiD on the Fama–French daily UMD factor (in basis points), parallel to the fund Panel B specification. Sample is restricted to treated days $\tau-4$ and $\tau-3$ and early-month controls $[\tau+5, \tau+8]$, 1980–2025, excluding May 2024 to avoid contamination at the regime boundary. T1Reform_t equals one for dates from June 1, 2024 onward, after the T+1 settlement reform took effect. Heteroskedasticity-robust standard errors in parentheses. A positive DiD coefficient indicates that UMD’s PreTOM concentration observed historically on $\tau-4$ has shifted to $\tau-3$ following the May 28, 2024 transition to T+1 settlement, mirroring the stock-level shift documented in Table 7 and the fund-level shift documented in Panel B below. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.12: Settlement Change Difference-in-Differences: UMD and Loser Funds (continued)

Panel B: $T+1$ settlement DiD on D1 fund portfolio (CRSP daily, 1998–2025)

	Simple	With FF3
Intercept	−3.70 (−0.96)	−2.50 (−1.43)
T1Reform _{<i>t</i>}	19.00 (1.15)	−1.25 (−0.26)
$\mathbf{1}_{[\tau-4]}$	0.09 (0.01)	0.23 (0.06)
$\theta_{\tau-4}$ (T1Reform _{<i>t</i>} × $\mathbf{1}_{[\tau-4]}$)	12.22 (0.49)	6.27 (0.57)
$\mathbf{1}_{[\tau-3]}$	19.80** (2.36)	0.33 (0.10)
$\theta_{\tau-3}$ (T1Reform _{<i>t</i>} × $\mathbf{1}_{[\tau-3]}$)	−39.36 (−1.64)	−23.64** (−2.40)
DiD ($\theta_{\tau-4} - \theta_{\tau-3}$)	51.58** (2.03)	29.91** (2.28)
Observations	1,959	1,959

Notes. Portfolio-level analogue of equation (7) applied to the equal-weighted D1 (loser-decile) fund portfolio, 1998–2025. Treated days are $\tau-4$ and $\tau-3$ relative to month-end; the control is early-month days $[\tau+5, \tau+8]$ (matching Table 7). T1Reform_{*t*} equals one for dates from June 1, 2024 onward, after the $T+1$ settlement reform took effect; May 2024 is excluded from both pre- and post-samples to avoid contamination at the regime boundary. Standard errors clustered by date are reported in parentheses. The DiD coefficient $\theta_{\tau-4} - \theta_{\tau-3}$ measures the predicted one-day shift in the selling window from $\tau-4$ to $\tau-3$ following the reform. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.15 Beta Controls

This section reports the beta-controlled regressions referenced in the main text. The concern is that the Loser $\times$ PreTOM effect could be a mechanical transmission of the aggregate market’s own PreTOM underperformance through higher loser betas, rather than targeted institutional selling of losers. We use 60-month rolling CAPM betas from the Jensen et al. (2023) global factor data, centered at the cross-sectional average beta on each date (approximately one, the market beta). The centering does not affect the estimates; it makes the coefficients easier to interpret: the Loser $\times$ PreTOM coefficient measures the PreTOM effect for a loser with an ordinary, market-like beta, which keeps it directly comparable to the baseline estimate in Table 2. The full specification adds β , PreTOM $\times\beta$, Loser $\times\beta$, and Loser $\times$ PreTOM $\times\beta$ to the baseline. The triple interaction, a slope with respect to beta, is unaffected by the centering and is the coefficient that would capture beta-amplification of loser selling if the channel were mechanical.

Tables IA.13, IA.14, and IA.15 show the results for value-weighted returns, equal-weighted returns, and TAQ net selling pressure respectively. Column 1 is the baseline; column 2 adds the PreTOM $\times\beta$ interaction in isolation to show its marginal contribution; column 3 is the saturated specification with the standalone β and all lower-order interactions included. The cleanest test of a beta-amplified loser-selling channel is the triple interaction Loser $\times$ PreTOM $\times\beta$ in column 3: it is statistically indistinguishable from zero in all three tables (for example, in the TAQ table, +0.0003 with $t = 0.48$). The headline Loser $\times$ PreTOM coefficient in the saturated specification is essentially unchanged from the baseline (TAQ: +0.0018** in column 3 versus +0.0020** in column 1). The marginal-contribution column 2 looks different from this picture only because, with no other beta terms in the regression, the standalone PreTOM $\times\beta$ absorbs part of the joint loser-PreTOM effect; once β , Loser $\times\beta$, and Loser $\times$ PreTOM $\times\beta$ are also included in column 3, PreTOM $\times\beta$ reverts to a small negative coefficient and the original Loser $\times$ PreTOM coefficient returns. The interpretation is that high-beta stocks mechanically track the market’s PreTOM drop in prices, but are not sold harder than other stocks on net during PreTOM; the loser-selling pattern lives on top of, not through, the market’s own calendar structure.

Table IA.13: Adding Stock-Level Beta as a Control to Table 2 (VW)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	2.609** (2.04)	2.772** (2.13)	2.622** (2.30)
Loser $\times$ PreTOM	-7.151*** (-3.08)	-6.488*** (-2.99)	-7.024*** (-3.55)
Beta			0.905 (0.91)
PreTOM $\times$ Beta		-2.694 (-1.62)	-3.693** (-2.00)
Loser $\times$ Beta			0.239 (0.17)
Loser $\times$ PreTOM $\times$ Beta			3.050 (1.36)
Fixed effects	Stock, Date	Stock, Date	Stock, Date
Observations	53,324,145	44,638,263	44,638,263
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable matches Table 2: the daily stock return in excess of the value-weighted CRSP market return ($\text{ExRet}_{i,t} = r_{i,t} - r_t^m$), in basis points. Returns are value-weighted (lagged market-cap). All specifications include stock and date fixed effects. t -statistics are in parentheses; standard errors are two-way clustered by stock and date. Beta is the JKP 60-month rolling CAPM beta, demeaned within each date using value weights. Sample: CRSP common stocks, 1980–2025; columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.14: Adding Stock-Level Beta as a Control to Table 2 (EW)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	8.022*** (11.19)	8.014*** (11.11)	8.008*** (11.93)
Loser $\times$ PreTOM	-2.550* (-1.85)	-2.471* (-1.89)	-2.331* (-1.85)
Beta			1.318** (2.43)
PreTOM $\times$ Beta		-3.222*** (-3.27)	-3.905*** (-3.34)
Loser $\times$ Beta			-0.520 (-0.74)
Loser $\times$ PreTOM $\times$ Beta			0.203 (0.16)
Fixed effects	Stock, Date	Stock, Date	Stock, Date
Observations	53,324,145	44,638,263	44,638,263
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable matches Table 2: the daily stock return in excess of the value-weighted CRSP market return ($\text{ExRet}_{i,t} = r_{i,t} - r_t^m$), in basis points. Equal-weighted across stocks. All specifications include stock and date fixed effects. t -statistics are in parentheses; standard errors are two-way clustered by stock and date. Beta is the JKP 60-month rolling CAPM beta, demeaned within each date (equal-weighted). Sample: CRSP common stocks, 1980–2025; columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.15: Beta as Control in TAQ Selling Pressure (Table 5)

	(1) Baseline	(2) + PreTOM $\times$ Beta	(3) Full
Loser	0.0122*** (20.20)	0.0126*** (19.64)	0.0121*** (17.83)
Loser $\times$ PreTOM	0.0020** (2.50)	0.0011 (1.40)	0.0018** (2.17)
Beta			0.0047*** (7.88)
PreTOM $\times$ Beta		0.0018*** (3.98)	-0.0007 (-1.51)
Loser $\times$ Beta			0.0005 (0.88)
Loser $\times$ PreTOM $\times$ Beta			0.0003 (0.48)
Fixed effects	Stock, Date	Stock, Date	Stock, Date
Observations	17,861,276	15,972,786	15,972,786
Within R^2	0.0002	0.0002	0.0002

Notes. Dependent variable matches Table 5: TAQ net selling pressure (sell volume minus buy volume, divided by total volume), which by construction ranges from -1 to $+1$; higher values indicate more selling pressure. All specifications include stock and date fixed effects. t -statistics are in parentheses; standard errors are two-way clustered by stock and date. Beta is the JKP 60-month rolling CAPM beta, demeaned within each date (equal-weighted). Sample: CRSP/TAQ stocks, 2003–2022; columns 2–3 restrict to the subsample with non-missing beta. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.16 European T+3-to-T+2 Settlement Transition

This section reports country-level evidence from the synchronized European migration from T+3 to T+2 settlement on October 6, 2014, implemented under the EU Central Securities Depositories Regulation (CSDR). The settlement mechanism predicts that this reform should move loser-selling pressure from $\tau-5$ to $\tau-4$. We estimate this boundary-day contrast separately by country and aggregate the estimates across countries.

Before turning to the difference-in-differences estimates, Figure IA.9 establishes that the transition markets share the U.S. loser-selling window. Pooled across the fifteen markets, mean loser excess returns (loser return minus the U.S. one-month T-bill rate, the same definition as the U.S. figure) decline from mid-month into a trough just before month-end and recover at month-end.⁷ Over the full sample (Panel A), the trough falls at the $[\tau-9, \tau-4]$ window and the immediately adjacent windows, which are statistically indistinguishable from one another, matching the location of the U.S. trough (Figure IA.2). In the pre-reform T+3 period (Panel B), the trough falls exactly at $[\tau-9, \tau-4]$, so the window that anchors the boundary-day comparison is identified by data that predate the reform. Adding the post-reform T+2 era (Panel A versus Panel B) shifts mass in the trough one day toward month-end, the direction the settlement mechanism predicts; we treat this comparison as suggestive, since the one-day shift is identified by the difference-in-differences estimates below rather than by trough locations across eras.

For each country c , we estimate equation (9) from the main text. The dependent variable is the value-weighted D1 loser portfolio return in excess of the country market return, measured in basis points. The treated days are $\tau-5$ and $\tau-4$. The omitted reference category is a mid-month control window $[\tau-15, \tau-10]$. The country-level contrast is $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$, where a positive value indicates that the old boundary day $\tau-5$ improves relative to the new boundary day $\tau-4$ after the reform.

Table IA.16 reports country-level estimates for the markets that transitioned, and Figure IA.10 displays the same estimates as a forest plot. Twelve of the fifteen country-level contrasts are positive, and the inverse-variance-weighted pooled estimate excludes zero, while

⁷Following the convention of Kenneth French's international factor files, the risk-free rate is the U.S. one-month T-bill rate. The risk-free rate is essentially constant within a month, so the within-month pattern, and in particular the location of the trough, is insensitive to this choice: using raw local-currency returns instead places the trough at the same $[\tau-9, \tau-4]$ window (untabulated).

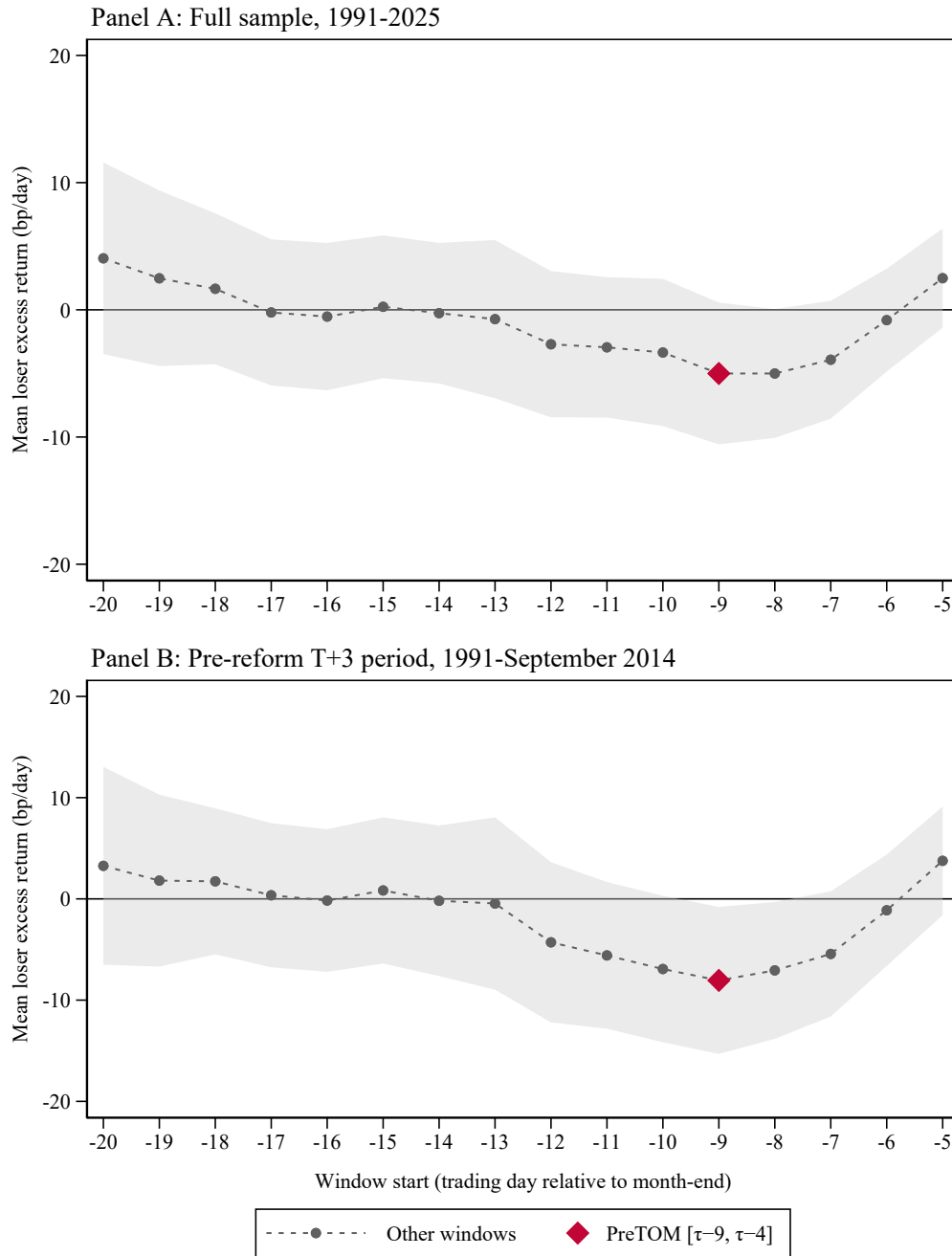


Figure IA.9: Sliding six-day window analysis for the European transition markets. Each point plots the mean daily VW loser excess return (loser return minus the U.S. one-month T-bill rate) for a six-day window starting at the indicated trading day, pooled across the fifteen European T+3-to-T+2 transition markets (equal-weighted within month; country-months with all six days present). Panel A uses the full sample (1991–2025); Panel B uses the pre-reform T+3 period (1991–September 2014). The diamond marks the PreTOM window $[\tau-9, \tau-4]$. Shaded areas are 95% confidence intervals from Newey-West standard errors (5 lags) on monthly means.

every individual country's confidence interval is wide: the identification comes from the agreement of the sign across independent markets rather than from any single country. The two negative non-trivial estimates, Switzerland and Ireland, carry the widest confidence intervals; Switzerland is the only market outside the EU and the CSDR scope.

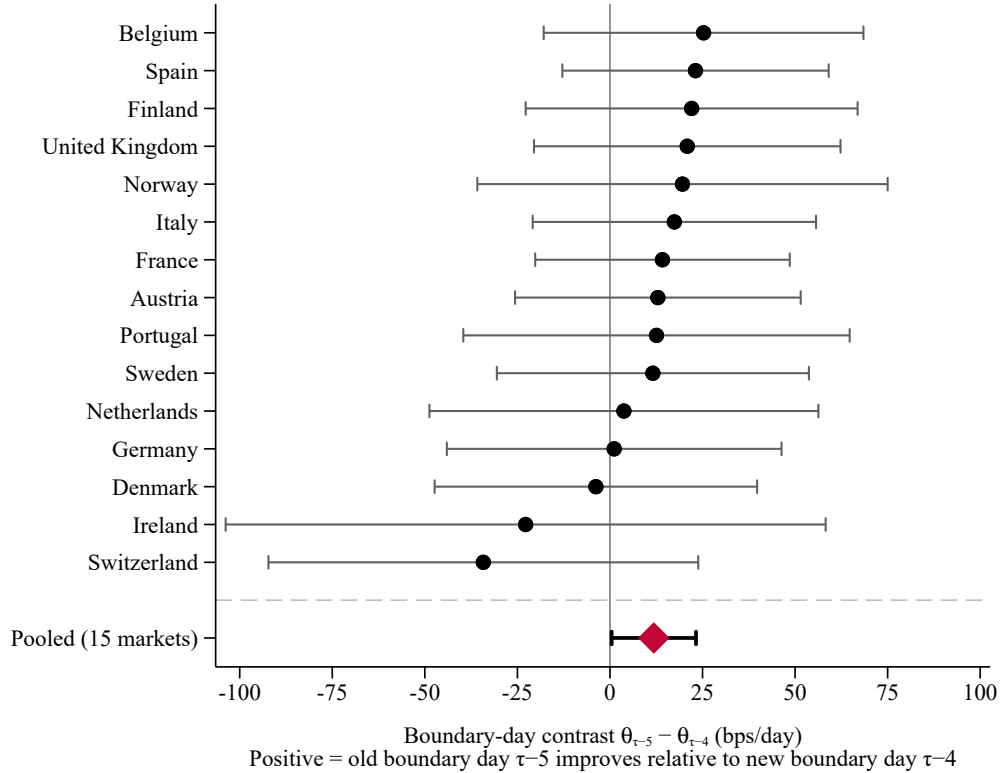


Figure IA.10: Country-level boundary-day contrasts for the European T+3-to-T+2 transition. Each dot is one country's difference-in-differences contrast $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$ from equation (9), the improvement of the old boundary day $\tau-5$ relative to the new boundary day $\tau-4$ after the October 2014 reform, with 95% confidence intervals from date-clustered standard errors. The diamond is the inverse-variance-weighted pooled estimate across the fifteen markets. Estimates are identical to Table IA.16.

Table IA.17 aggregates the country-level contrasts. The full synchronized European sample yields a positive pooled contrast, and the random-effects (DerSimonian-Laird) variant is identical to the fixed-effect estimate because the estimated between-country heterogeneity is zero. Date-clustered panel specifications are reported as stress tests because the reform date

is common across markets; these specifications are less informative about country-level sign concordance but useful for assessing sensitivity to common date shocks. Table IA.18 reports leave-one-country-out estimates. The pooled estimate remains positive after dropping any single country, ranging from +10.59 to +13.71 bps.

Table IA.16: Country-Level Boundary Contrasts: European T+3-to-T+2 Transition

Market	Contrast κ_c (bps)	SE (bps)	t	N obs (days)	Predicted sign?	Sample flag
Austria (AUT)	+12.94	19.68	+0.66	3,089	+	full sample
Belgium (BEL)	+25.28	22.03	+1.15	3,230	+	full sample
Switzerland (CHE) [†]	−34.20	29.61	−1.16	3,016	−	outside CSDR scope
Germany (DEU)	+1.14	23.06	+0.05	3,360	+	full sample
Denmark (DNK)	−3.80	22.22	−0.17	3,008	−	full sample
Spain (ESP)	+23.11	18.35	+1.26	3,360	+	full sample
Finland (FIN)	+22.06	22.87	+0.96	3,202	+	full sample
France (FRA)	+14.18	17.54	+0.81	3,360	+	full sample
United Kingdom (GBR)	+20.88	21.12	+0.99	3,355	+	full sample
Ireland (IRL)	−22.77	41.34	−0.55	2,974	−	full sample
Italy (ITA)	+17.40	19.52	+0.89	3,360	+	full sample
Netherlands (NLD)	+3.77	26.80	+0.14	3,325	+	full sample
Norway (NOR)	+19.59	28.27	+0.69	3,336	+	full sample
Portugal (PRT)	+12.57	26.62	+0.47	2,853	+	full sample
Sweden (SWE)	+11.59	21.50	+0.54	3,280	+	full sample

Notes. Per-country DiD contrasts $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$ from equation (9). The treated days are $\tau-5$ and $\tau-4$ (τ is the last trading day of the month); the omitted reference is the mid-month window $[\tau-15, \tau-10]$. The reform date is October 6, 2014. Standard errors are clustered by date. Positive estimates indicate that the old boundary day $\tau-5$ improves relative to the new boundary day $\tau-4$ after the reform. [†]Switzerland is outside the EU and outside the scope of the Central Securities Depositories Regulation (CSDR), but the SIX Swiss Exchange transitioned to T+2 on the same date; we report it for completeness but exclude it from the “ex-Switzerland” aggregation in Table IA.17.

Table IA.17: Meta-Analytic Aggregation: European T+3-to-T+2 Boundary Contrasts

Specification	Pooled contrast	SE	t	p	Sign (positive)	Sign-test p (one-sided)
<i>Panel A: Country as the replication unit (inverse-variance-weighted)</i>						
Full European sample (15 markets)	+11.85	5.82	+2.04	0.042	12/15	0.018
Ex Switzerland (14)	+13.71	5.94	+2.31	0.021	12/14	0.006
<i>Panel B: Robustness aggregations</i>						
Unweighted country mean (15)	+8.25	4.44	+1.86	0.084	12/15	—
Random-effects (DL), 15	+11.85	5.82	+2.04	0.042	—	—
<i>Panel C: Date-clustered panel stress tests</i>						
Pooled, date-cluster, 15 markets	+8.90	11.00	+0.81	0.418	—	—

Notes. The table aggregates the country-level contrasts $\kappa_c = \theta_{c,\tau-5} - \theta_{c,\tau-4}$ reported in Table IA.16. Panel A pools by inverse-variance weighting, treating each country as a replication unit. Sign-test p -values are exact binomial p -values under the null of equal probability of positive and negative signs. Panel B reports the unweighted country mean (with cross-country standard error of the mean) and the random-effects DerSimonian-Laird variant; the latter coincides with the fixed-effect estimate because the estimated between-country heterogeneity is zero. Panel C reports a panel specification pooling all country-day observations with country fixed effects and standard errors clustered by date; this specification is conservative against common-date shocks but does not exploit the cross-country sign concordance documented in Panel A. “Ex Switzerland” excludes Switzerland, which is outside the EU and outside the scope of the Central Securities Depositories Regulation (CSDR).

Table IA.18: Leave-One-Country-Out Robustness: European T+3-to-T+2 Pool

Dropped country	Pooled κ	SE	t	p	Sign (positive)
Austria (AUT)	+11.75	6.10	+1.93	0.054	11/14
Belgium (BEL)	+10.84	6.04	+1.80	0.072	11/14
Switzerland (CHE)	+13.71	5.94	+2.31	0.021	12/14
Germany (DEU)	+12.58	6.02	+2.09	0.037	11/14
Denmark (DNK)	+13.01	6.03	+2.16	0.031	12/14
Spain (ESP)	+10.59	6.14	+1.73	0.085	11/14
Finland (FIN)	+11.15	6.02	+1.85	0.064	11/14
France (FRA)	+11.57	6.17	+1.87	0.061	11/14
UK (GBR)	+11.11	6.06	+1.83	0.067	11/14
Ireland (IRL)	+12.55	5.88	+2.13	0.033	12/14
Italy (ITA)	+11.31	6.10	+1.85	0.064	11/14
Netherlands (NLD)	+12.25	5.97	+2.05	0.040	11/14
Norway (NOR)	+11.51	5.95	+1.93	0.053	11/14
Portugal (PRT)	+11.82	5.97	+1.98	0.048	11/14
Sweden (SWE)	+11.87	6.05	+1.96	0.050	11/14

Notes. Each row reports the inverse-variance-weighted pooled estimate after dropping the listed country from the fifteen-market sample. The pooled estimate remains positive in every leave-one-out specification, ranging from +10.59 to +13.71 basis points per day. The two largest reductions (dropping ESP and BEL) leave the pooled estimate at +10.6 and +10.8 bps respectively, both with t between 1.7 and 1.8.

IA.17 U.S. T+3-to-T+2 Settlement Transition

An earlier U.S. settlement reform, the September 5, 2017 transition from T+3 to T+2 equity settlement, provides a directional check on the same comparative static. The dash-for-cash mechanism predicts a one-day shift in loser-selling pressure at this reform, from $\tau-5$ (the T+3-era boundary day) to $\tau-4$ (the T+2-era boundary day).

Table IA.19 reports the deadline-day DiD using the full T+3 regime as the pre-period (June 1995 – August 2017) and the T+2 era as the post-period (October 2017 – April 2024, excluding both September 2017, the regime-boundary month, and the T+1 era from May 2024 onward). The $\tau-5$ vs. $\tau-4$ DiD is +13.08 basis points ($F = 0.27$, $p = 0.60$): the point estimate is in the predicted direction (selling pressure moves toward $\tau-4$) but is not statistically distinguishable from zero.

Table IA.20 reports unconditional era-means of the boundary spread $R_{\tau-5}^{D1} - R_{\tau-4}^{D1}$, which communicate the same directional pattern as the DiD. In the pre-T+2 era (January 1980 – August 2017), the spread averages -1.19 basis points per day (with $\tau-5$ slightly worse than $\tau-4$). In the T+2 era (September 2017 – April 2024), the spread averages $+13.36$ basis points per day, with $\tau-4$ becoming the worse boundary day. The T+1 era (June 2024 – December 2025) has only nineteen months of post-reform observations and is reported for completeness.

Aggregate mutual fund flows provide independent evidence of the same one-day shift. Table IA.21 reports mean daily net flows into U.S. mutual funds, aggregated across all funds in the Morningstar daily flow data, at each trading day from $\tau-9$ to $\tau-4$, separately for the pre-T+2 era (January 2007 – August 2017) and the T+2 era (October 2017 – June 2023). Under T+3 settlement, $\tau-4$ lies past the marginal cash-raising day and shows net inflows ($+0.56$ basis points of assets per day). Under T+2, $\tau-4$ becomes the marginal cash-raising day, and flows on that day flip to net outflows (-0.51 basis points per day; difference $t = -1.89$). Outflows over $[\tau-7, \tau-5]$ also intensify after the reform. The flow data thus show the movement of month-end liquidity demand one day closer to month-end than the return-based DiD is underpowered to detect.

Table IA.19: T+3→T+2 Settlement Boundary-Day Difference-in-Differences
Panel A: 2×2 Cell Means (loser market-adjusted return, bps/day)

	Pre-T+2 (Jun 1995–Aug 2017)	Post-T+2 (Oct 2017–Apr 2024)	Diff
$T-5$	−6.09 ($N = 267$)	−11.26 ($N = 79$)	−5.16
$T-4$	−6.86 ($N = 267$)	−25.10 ($N = 79$)	−18.24
Diff ($T-5$ minus $T-4$)	0.77	13.85	13.08

Panel B: DiD Regression with Early-Month Control

	Coefficient	Robust SE
θ_{T-5} : $\mathbf{1}_{[T-5]} \times \text{Post } T+2$	−7.92	(21.32)
θ_{T-4} : $\mathbf{1}_{[T-4]} \times \text{Post } T+2$	−21.00	(21.73)
$\theta_{T-5} - \theta_{T-4}$ (boundary shift)	13.08	(25.22)
$F = 0.27, p = 0.604$		

Notes: Boundary-day difference-in-differences test of the T+3 to T+2 settlement transition (September 5, 2017). Dependent variable: daily market-adjusted VW loser-portfolio return (basis points). Treated days are $T-5$ and $T-4$ (the pre- and post-reform boundary days under the linearity prediction); control days are early-month $[T+5, T+8]$. Pre-period: June 1995 (start of the T+3 regulatory regime) to August 2017; post-period: October 2017 to April 2024 (excluding both September 2017, the regime-boundary month, and the T+1 era from May 2024 onward). The headline test $\theta_{T-5} - \theta_{T-4} > 0$ is in the predicted direction (parallel to the T+1 test in the main paper: the old boundary day $T-5$ improves and the new boundary day $T-4$ worsens), but not statistically distinguishable from zero. Robust standard errors in parentheses. By construction, the Panel A $T-5$ minus $T-4$ diff equals the Panel B $\theta_{T-5} - \theta_{T-4}$; the early-month control absorbs the common post-period drift but cancels out of the boundary contrast.

Table IA.20: Era-Means of the U.S. Boundary Spread $R_{\tau-5}^{D1} - R_{\tau-4}^{D1}$

Era	Spread (bps/day)	n months	Interpretation
Pre-T+2 (Jan 1980 – Aug 2017)	−1.19	452	$\tau-5$ slightly worse than $\tau-4$
T+2 era (Sep 2017 – Apr 2024)	+13.36	80	$\tau-4$ becomes the worse boundary day
T+1 era (Jun 2024 – Dec 2025)	−62.93	19	$\tau-4$ no longer a boundary day

Notes. Era-means of the unconditional monthly boundary spread, defined as the loser-portfolio market-adjusted return at $\tau-5$ minus the corresponding return at $\tau-4$ (τ is the last trading day of the month). Positive values indicate $\tau-4$ is the more negative day, consistent with $\tau-4$ being the binding cash-raising boundary. The T+1 era begins in June 2024: the May 28, 2024 reform postdates the $\tau-5/\tau-4$ boundary days of May 2024, which therefore remain in the T+2 selling regime and are excluded, matching the main-text post-reform convention. The T+1-era row has only nineteen months of post-reform observations and is reported for completeness. Under T+1 settlement, the cash-raising boundary shifts to $\tau-3$ (see main-text Section 5.1), so $\tau-4$ ceases to be a boundary day in that era and the $\tau-5$ vs. $\tau-4$ comparison is no longer informative about the settlement mechanism for the T+1 regime.

Table IA.21: Aggregate Mutual Fund Flows Around the T+3-to-T+2 Settlement Transition

Daily net flow (bps of assets)	$\tau-9$	$\tau-8$	$\tau-7$	$\tau-6$	$\tau-5$	$\tau-4$
Pre-T+2 era (Jan 2007 – Aug 2017)	−0.37	−0.23	−0.26	−0.10	−0.07	+0.56
T+2 era (Oct 2017 – Jun 2023)	−0.04	+0.09	−0.57	−0.66	−0.60	−0.51
Difference	+0.33 (0.37)	+0.32 (0.36)	−0.31 (−0.41)	−0.56 (−0.71)	−0.53 (−0.70)	−1.07 (−1.89)

Notes. Mean daily aggregate net flows into U.S. mutual funds at each trading day relative to month-end (τ is the last trading day), in basis points of aggregate fund assets per day. Flows are aggregated across all funds in the Morningstar daily flow data, which begins in January 2007. The pre-T+2 era covers January 2007 through August 2017 ($n = 128$ months); the T+2 era covers October 2017 through June 2023, the end of the flow data ($n = 69$ months). September 2017, the regime-boundary month, is excluded. The bottom rows report the T+2-era mean minus the pre-T+2-era mean and, in parentheses, the t -statistic from a two-sample test of the difference. Under T+3 settlement the marginal cash-raising day is $\tau-5$ and $\tau-4$ shows net inflows; under T+2 the marginal cash-raising day moves to $\tau-4$, which flips to net outflows.

IA.18 Winner Window Effect

The asymmetry of the PreTOM effect, mostly loser-driven, is confirmed by running the baseline regression on winners instead of losers. Table IA.22 reports the winner analogue of the baseline Loser $\times$ PreTOM regression. Neither the Winner main effect nor the Winner $\times$ PreTOM interaction is statistically distinguishable from zero in the value-weighted specification, confirming that winners show no differential PreTOM concentration.

Table IA.22: Winner Window Effect on Daily Returns

	(1) EW	(2) VW
Winner	−0.23*** (−3.29)	0.04 (0.40)
Winner $\times$ PreTOM	−1.27 (−0.82)	1.55 (0.76)
Fixed effects	Stock, Date	Stock, Date
Observations	53,324,149	53,324,149
Within R^2	0.0000	0.0000

Notes. Dependent variable is the daily stock return in excess of the value-weighted CRSP market ($r_{i,t} - r_t^m$), in basis points. Value-weighted (VW) regressions use lagged market-capitalization weights. t -statistics (in parentheses) are computed from standard errors double-clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: CRSP common stocks, 1980–2025.

IA.19 S&P 500 Membership Robustness

A natural concern is that the PreTOM effect is concentrated among small-cap or thinly traded stocks. The effect is also present among S&P 500 constituents: the Loser $\times$ PreTOM coefficient is -8.77 ($t = -2.71$) among S&P 500 stocks and -4.62 ($t = -2.53$) among non-constituents, with no statistically significant difference between the two groups (triple interaction $t = -0.82$). The effect is not confined to small or illiquid stocks.

Table IA.23: PreTOM Effect by S&P 500 Membership

	Non-S&P 500 (1)	S&P 500 (2)	Full Sample (3)
Loser	1.806* (1.84)	4.018** (2.23)	2.429** (2.20)
Loser $\times$ PreTOM	-4.622^{**} (-2.53)	-8.769^{***} (-2.71)	-5.916^{**} (-2.56)
Loser $\times$ S&P 500			0.408 (0.25)
Loser $\times$ PreTOM $\times$ S&P 500			-2.520 (-0.82)
Fixed effects	Stock, Date	Stock, Date	Stock, Date
Observations	47,863,849	5,460,280	53,324,145
Within R^2	0.0000	0.0000	0.0000

Notes. Dependent variable is the daily stock return in excess of the value-weighted CRSP market ($r_{i,t} - r_t^m$), in basis points. Value-weighted using lagged market-capitalization weights. Columns (1) and (2) restrict the sample to non-S&P 500 and S&P 500 constituent stocks, respectively; columns (1)+(2) do not exactly sum to column (3) because the S&P 500 membership flag is missing on a small number of stock-days (16 obs), which are dropped from the split samples but retained in the full-sample interaction specification. Column (3) estimates the full sample with an S&P 500 interaction. All specifications include stock and date fixed effects. t -statistics (in parentheses) are computed from standard errors two-way clustered by stock and date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sample: CRSP common stocks, 1980–2025.

IA.20 Long-Horizon Reversion: Identification of the PreTOM-Specific Component

The short-horizon Post-window narrowing documented in the main paper shows that the loser-vs-non-loser gap that opens during PreTOM closes within the surrounding seven days. The same mechanism implies a longer-horizon prediction: cumulative pressure from past PreTOM windows should mean-revert at horizons of several months to two years, while cumulative returns over non-PreTOM days, which reflect information and fundamentals rather than transitory flow pressure, should not. The asymmetric prediction tightens the test: identification comes from the contrast between the two predictors rather than the absolute level of either.

We focus on the D1 loser decile. Panel A of Table IA.24 reports daily time-series regressions of the value-weighted D1 portfolio return r_t on its own cumulative past returns:

$$r_t = \alpha + \beta_1 \text{CumPT}_t + \beta_2 \text{CumNPT}_t + \beta_3 \text{CumRec}_t + \beta_4 \text{PreTOM}_t + \beta_5 r_{t-1} + \varepsilon_t, \quad (\text{IA.3})$$

where $\text{CumPT}_t = \sum_{s=t-480}^{t-60} r_s \cdot \mathbf{1}\{s \in \text{PreTOM}\}$ sums distant past returns on PreTOM days only; $\text{CumNPT}_t = \sum_{s=t-480}^{t-60} r_s \cdot \mathbf{1}\{s \notin \text{PreTOM}\}$ sums distant past returns on non-PreTOM days; and $\text{CumRec}_t = \sum_{s=t-60}^{t-1} r_s$ is the recent 60-day cumulative return. The distant window runs from 60 to 480 trading days prior (approximately three to twenty-four months). The regression is estimated with Newey–West standard errors at 21 lags. The prediction is $\beta_1 < 0$ and $\beta_2 \approx 0$, with the Wald test of $\beta_1 = \beta_2$ rejecting symmetric reversion.

The pattern holds in the standard D1 portfolio ($\beta_1 = -0.0034$, $t = -2.37$; Wald $t = -2.33$) and strengthens when the portfolio is restricted to stocks with persistent past D1-PreTOM exposure. Conditioning today’s portfolio on stocks that were in D1 on at least three past PreTOM days (out of approximately 120 in the distant window) drives β_1 to -0.0043 ($t = -3.06$) and the Wald statistic to $t = -3.04$. The filter keeps 575 of the 875 stocks in the average unfiltered D1 cross-section, holding 55 percent of D1 market capitalization; the removed stocks are recent D1 entrants with little or no past D1-PreTOM exposure, for which accumulated past PreTOM pressure is not defined. Results are stable across thresholds from ≥ 1 through ≥ 30 ; we report ≥ 3 as the primary specification. Within the ≥ 3 universe, CumPT_t has a time-series standard deviation of 1,779 basis points; the implied

cross-sectional spread between $\pm 1\sigma$ tails is 15.4 basis points per day, roughly 39 percentage points annualized. Conditioning on past D1 status leaves β_2 negligible in both specifications (0.0000 unfiltered, -0.0001 filtered), confirming the absence of conventional medium-horizon momentum in this window (Jegadeesh and Titman, 1993).

Panel B broadens the sample to all momentum deciles and identifies the same reversion within-stock, with stock and day fixed effects absorbing stock-specific means and aggregate day-level shocks. Broadening beyond D1 is necessary because stocks must transition in and out of D1 over time for the within-stock decomposition by past D1 status to be identified. The key step is to decompose each stock's past PreTOM returns by its own past D1 status: $\text{CumPT}_{i,t}^{\text{PL}}$ sums past PreTOM returns on days when stock i was itself in D1, and $\text{CumPT}_{i,t}^{\text{PNL}}$ sums past PreTOM returns from periods when the stock was not. Analogous quantities $\text{CumNPT}_{i,t}^{\text{PL}}$ and $\text{CumNPT}_{i,t}^{\text{PNL}}$ decompose past non-PreTOM returns. The panel regression is

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha_i + \delta_t + \beta_1 \text{CumPT}_{i,t}^{\text{PL}} + \beta_2 \text{CumPT}_{i,t}^{\text{PNL}} \\ & + \beta_3 \text{CumNPT}_{i,t}^{\text{PL}} + \beta_4 \text{CumNPT}_{i,t}^{\text{PNL}} + \beta_5 \text{CumRec}_{i,t} + \beta_6 \text{Loser}_{i,t} + \varepsilon_{i,t}, \end{aligned} \quad (\text{IA.4})$$

estimated on all momentum deciles (43.5 million stock-days), value-weighted by lagged market capitalization, with standard errors two-way clustered by stock and day.

Past PreTOM returns accumulated while the stock was in D1 ($\beta_1 = -0.00111$, $t = -4.99$) revert about three times more strongly than past PreTOM returns accumulated while the stock was outside D1 ($\beta_2 = -0.00038$, $t = -2.65$), with the Wald test of equality rejecting at $p = 0.0006$. Past non-PreTOM returns show no equivalent asymmetry: β_3 is indistinguishable from zero in D1 and β_4 is marginal outside D1. The marginal β_4 is consistent with a small generic long-horizon reversal in non-loser stocks that stock fixed effects do not fully absorb; its magnitude is about 60 percent of β_2 .

If the negative coefficient on CumPT^{PL} reflected generic mean-reversion of persistent underperformers, the analogous $\text{CumNPT}^{\text{PL}}$ should also be negative and large; it is not. The reversion loads specifically on the component of past returns that accumulated during PreTOM windows, when flow pressure was active. Standard persistent-regressor concerns (Stambaugh, 1999; Ferson et al., 2003) affect both predictors similarly given their parallel construction and difference out in the Wald test.

Panel B identifies whether each predictor matters; it does not say where the reversion shows up. Since the mechanism in the main paper concentrates on stocks currently classified as losers, we ask whether reversion of past D1-PreTOM accumulation realizes when stock i remains in D1 today, when it has exited, or both. We augment equation (IA.4) by interacting all four cumulative-return predictors with $\text{Loser}_{i,t}$:

$$\begin{aligned} r_{i,t} - r_t^m = & \alpha_i + \delta_t + \sum_{k=1}^4 \beta_k X_{i,t}^{(k)} + \sum_{k=1}^4 \delta_k X_{i,t}^{(k)} \times \text{Loser}_{i,t} \\ & + \beta_5 \text{CumRec}_{i,t} + \beta_6 \text{Loser}_{i,t} + \beta_7 \text{PreTOM}_t + \varepsilon_{i,t}, \end{aligned} \quad (\text{IA.5})$$

where $X_{i,t}^{(1)} = \text{CumPT}_{i,t}^{\text{PL}}$, $X_{i,t}^{(2)} = \text{CumPT}_{i,t}^{\text{PNL}}$, $X_{i,t}^{(3)} = \text{CumNPT}_{i,t}^{\text{PL}}$, and $X_{i,t}^{(4)} = \text{CumNPT}_{i,t}^{\text{PNL}}$. The main coefficients β_k now capture reversion on stock-days where i is currently outside D1; the linear combinations $\beta_k + \delta_k$ capture reversion on stock-days where i remains in D1. Panel C of Table IA.24 reports the results.

Among stocks currently outside D1, only past D1-PreTOM accumulation predicts future returns ($\beta_1 = -0.000713$, $t = -3.61$); the other three components are indistinguishable from zero. This is the post-exit prediction the main paper's narrative requires: once a stock leaves D1, only its history of having been pressed during PreTOM, and only on PreTOM days while it was in D1, predicts its subsequent path. In-D1 stock-days show stronger reversion across all four components ($\beta_k + \delta_k$ negative and statistically significant for $k = 1, \dots, 4$), but the all-negative pattern combines reversion with the mechanical contrarian property of the loser portfolio: stocks classified as losers today are stocks that have just fallen, and the sample is dense in further short-horizon mean-reversion of recent losses (Lehmann, 1990; Jegadeesh and Titman, 1995). The relevant test is therefore the out-of-D1 column, where this density is absent. There, CumPT^{PL} stands alone.

Table IA.24: Long-Horizon Reversion of PreTOM Loser Pressure

<i>Panel A: Portfolio Time Series (D1, value-weighted)</i>				
Portfolio	Avg stocks/day	β_1 : CumPT	β_2 : CumNPT	Wald: $\beta_1=\beta_2$
Standard D1 (no filter)	875	−0.0034** (−2.37)	0.0000 (−0.05)	$t = -2.33^{**}$
Persistent ≥ 3 (primary)	575	−0.0043*** (−3.06)	−0.0001 (−0.06)	$t = -3.04^{***}$
Persistent ≥ 30	399	−0.0036*** (−2.63)	0.0000 (−0.03)	$t = -2.43^{**}$
Persistent ≥ 110	35	−0.0019 (−1.44)	−0.0006 (−0.96)	$t = -0.90$

Notes (Panel A). Daily time-series regressions of value-weighted D1 portfolio returns on rolling cumulative past returns over the distant window $[t-480, t-60]$. CumPT_{*t*} sums returns on past PreTOM days ($\tau-9$ through $\tau-4$, where τ is the last trading day of the month); CumNPT_{*t*} sums returns on past non-PreTOM days. Specifications include a PreTOM_{*t*} dummy and the portfolio's own one-day lagged return as additional controls. Newey-West standard errors, 21 lags. *t*-statistics are in parentheses. The persistent- $\geq k$ portfolios restrict today's D1 universe to stocks that were also in D1 on at least *k* PreTOM days within the distant window $[t-480, t-60]$ (out of approximately 120 such days). The average number of stocks per day falls as the persistence filter tightens. Panels B and C continue overleaf. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table IA.24: Long-Horizon Reversion of PreTOM Loser Pressure (continued)

<i>Panel B: Stock-Level Panel (all deciles, stock and date fixed effects, VW)</i>			
Variable	Coefficient		<i>t</i> -stat
β_1 : CumPT $_{i,t}^{\text{PL}}$ (past PreTOM while in D1)	−0.00111***		(−4.99)
β_2 : CumPT $_{i,t}^{\text{PNL}}$ (past PreTOM while not in D1)	−0.00038***		(−2.65)
β_3 : CumNPT $_{i,t}^{\text{PL}}$ (past non-PreTOM while in D1)	−0.00013		(−0.79)
β_4 : CumNPT $_{i,t}^{\text{PNL}}$ (past non-PreTOM while not in D1)	−0.00023*		(−1.88)
β_5 : CumRec $_{i,t}$	−0.00253***		(−7.14)
β_6 : Loser $_{i,t}$	−2.16**		(−2.24)
Wald test: $\beta_1 = \beta_2$	$F = 11.85, p = 0.0006$		
Difference ($\beta_1 - \beta_2$)	−0.00073***		(−3.44)
Observations (stock-days)			43,463,404

<i>Panel C: Reversion by Current D1 Status (predictors interacted with Loser$_{i,t}$)</i>			
Past-return component	Out of D1 (main: β_k)	In D1 (main + int.: $\beta_k + \delta_k$)	× Loser (δ_k)
CumPT $_{i,t}^{\text{PL}}$	−0.000713*** (−3.61)	−0.001569*** (−4.33)	−0.000857** (−2.38)
CumPT $_{i,t}^{\text{PNL}}$	−0.000211 (−1.50)	−0.001484*** (−3.85)	−0.001273*** (−3.46)
CumNPT $_{i,t}^{\text{PL}}$	0.000116 (0.69)	−0.000574** (−2.13)	−0.000690*** (−2.59)
CumNPT $_{i,t}^{\text{PNL}}$	−0.000161 (−1.36)	−0.000720*** (−2.67)	−0.000559** (−2.27)
Wald test: $\delta_1 = 0$ (interaction on CumPT $^{\text{PL}}$)			$F = 5.67, p = 0.0173$
Joint test: $\delta_1 = \delta_2 = \delta_3 = \delta_4 = 0$			$F = 4.92, p = 0.0006$
Test $\beta_1 = 0$ (out-of-D1 reversion of CumPT $^{\text{PL}}$)			$F = 13.02, p = 0.0003$
Test $\beta_1 + \delta_1 = 0$ (in-D1 reversion of CumPT $^{\text{PL}}$)			$F = 18.79, p < 0.001$
Observations (stock-days)			43,463,404
Within R^2			0.2304

Notes (Panels B and C). Predictors are rolling cumulative returns over the distant window $[t-480, t-60]$, decomposed by whether stock i was itself in D1 on each past day: CumPT $^{\text{PL}}$ (CumPT $^{\text{PNL}}$) sums past PreTOM returns when stock i was (was not) in D1 on day s , and analogously for non-PreTOM days. CumRec $_t$ sums returns over the recent $[t-60, t-1]$ window. Panel B reports the panel regression of equation (IA.4); all deciles, stock and date FE, VW by lagged mcap, errors clustered by stock and date. Panel C augments by interacting all four cumulative-return predictors with Loser $_{i,t}$; “Out of D1” columns report the main coefficient β_k , “In D1” columns report $\beta_k + \delta_k$, rightmost column reports the interaction δ_k . The regression additionally controls for CumRec $_{i,t}$, Loser $_{i,t}$, and PreTOM $_t$. NYSE breakpoints (CRSP common stocks); deciles fixed within each calendar month. Sample 1980–2025. *t*-statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IA.21 Additional Figures

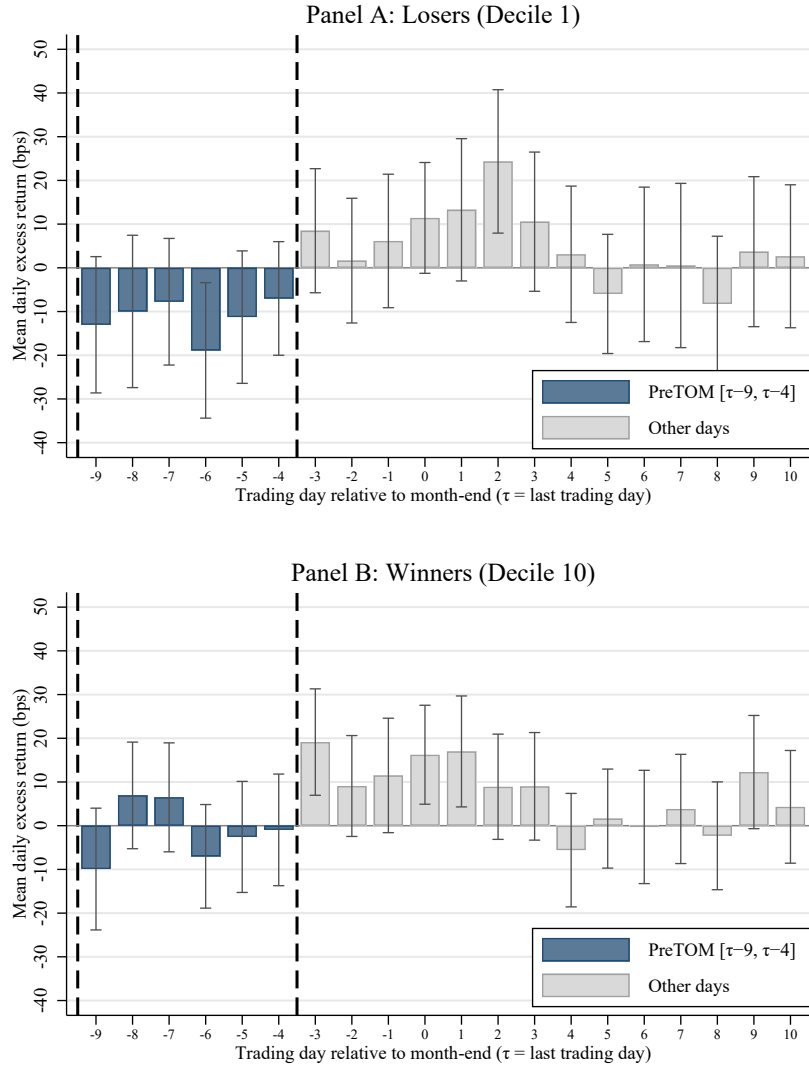


Figure IA.11: Mean daily excess returns ($r_{i,t} - r_t^f$, bps) by trading day relative to the turn-of-the-month (τ is the last trading day of the month). Panel A: D1 losers. Panel B: D10 winners. PreTOM bars ($[\tau-9, \tau-4]$) dark blue, other days light gray. Bars are the cross-month mean of the daily return at each trading day; whiskers are 95% confidence intervals. Value-weighted. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025.

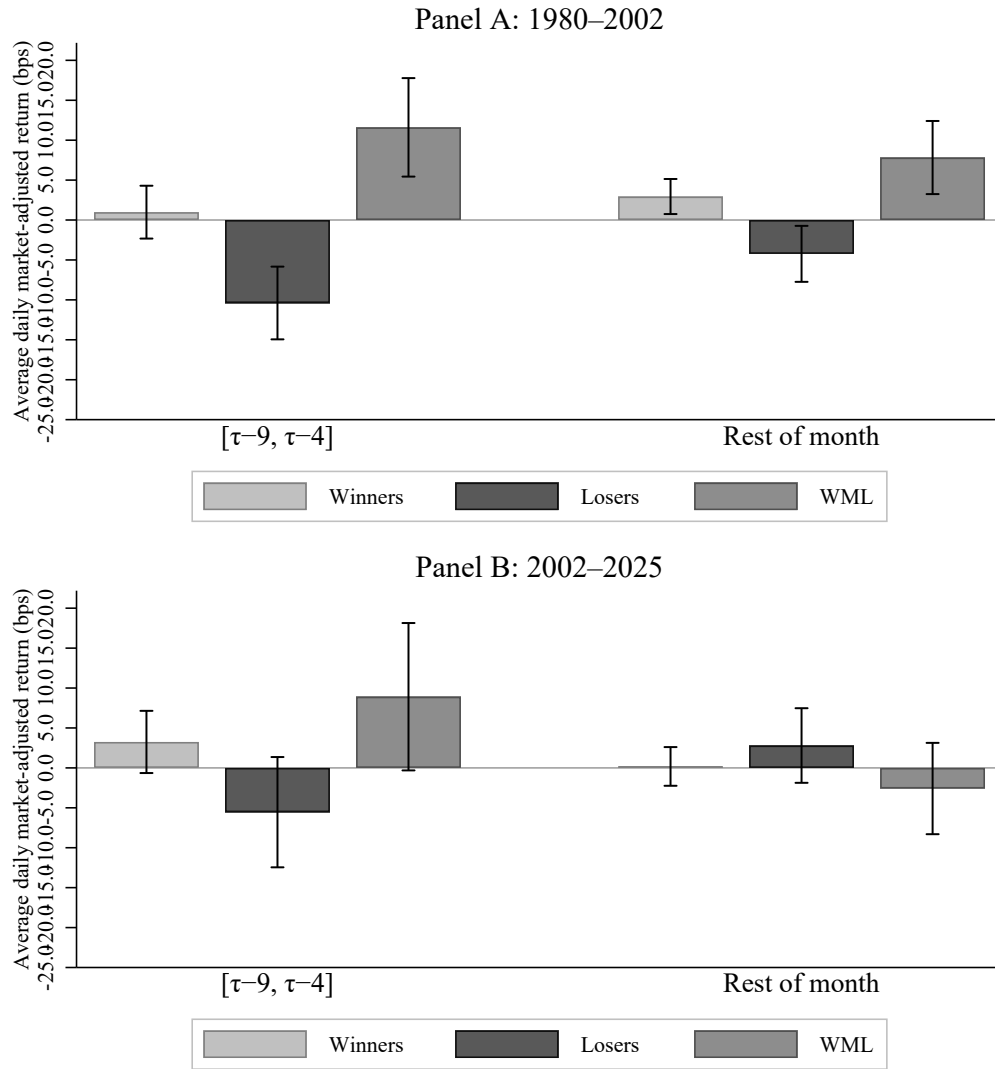


Figure IA.12: Average daily market-adjusted returns (basis points) of momentum winners, losers, and WML during PreTOM ($[\tau-9, \tau-4]$) versus Rest, by subperiod. Daily returns are compounded within each month-window and averaged across months; 95% confidence intervals use heteroskedasticity-robust standard errors. Value-weighted. Momentum deciles use NYSE breakpoints applied to all NYSE, AMEX, and NASDAQ stocks in CRSP; assignments are held constant within each calendar month. Sample: 1980–2025, split at mid-2002.

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